

JOHANNES GUTENBERG-UNIVERSITÄT, MAINZ

LECTURE NOTES ON

Topics on F-Singularities

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Foreword and Notation

These notes originate from the lecture course “Topics on F -singularities” given in the winter term 2024/2025 at the Johannes Gutenberg-University Mainz by Prof. Dr. Manuel Blickle.

1 The Frobenius and Regularity

The goal of this chapter is to give an introduction into positive characteristic geometry and to see how the Frobenius morphism can be used to measure the regularity of a ring. We begin with a reminder of the basics of positive characteristics commutative algebra.

For this whole chapter fix a prime number $p \in \mathbb{Z}$. We further assume all rings to be commutative and to have a unit and all ring homomorphism to be unitary — unless otherwise stated.

Definition 1.1. A ring A is called a $\mathbb{F}_p$ -Algebra, if the canonical morphism $\mathbb{Z} \rightarrow A$ factors as

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\quad} & A \\ \downarrow & \nearrow & \\ \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} & & \end{array}$$

where we use the standard notation $\hookrightarrow$ to denote a monomorphism.

Remark 1.2. $\mathbb{F}_p$ -algebras together with obvious commutative diagrams as morphisms form a category which we denote by $\text{Alg}_{\mathbb{F}_p}$. Note that this is nothing else then the slice category $\mathbb{F}_p/\text{CRing}$ of Rings under $\mathbb{F}_p$.

Definition and Proposition 1.3. Let $R \in \text{Alg}_{\mathbb{F}_p}$. Then the map

$$\begin{aligned} F : R &\rightarrow R \\ r &\mapsto r^p \end{aligned}$$

is a morphism in $\text{Alg}_{\mathbb{F}_p}$.

The proof of this is standard and left as an exercise to the reader. If we are concerned with multiple $R_i \in \text{Alg}_{\mathbb{F}_p}$ we sometimes write F_{R_i} to clarify the Ring we are talking about in the moment. Further we write F^e for the e -fold iteration of F .

We want to do geometry so lets consider the map induced by F on the affine F -scheme $\text{spec}(R)$. It is clear what happens on the structure sheaf so let us consider the map of topological spaces:

$$\begin{aligned} F : \text{Spec } R &\rightarrow \text{Spec } R \\ \mathfrak{p} &\mapsto F^{-1}(\mathfrak{p}) \end{aligned}$$

We have $\mathfrak{p} \subseteq F^{-1}(\mathfrak{p}) = \{r \in R \mid r^p \in \mathfrak{p}\} \subseteq \sqrt{\mathfrak{p}} = \mathfrak{p}$. Hence the Frobenius on the underlying topological space of an affine $\mathbb{F}_p$ -scheme is just the identity. This motivates the following

Definition 1.4. Let X be an $\mathbb{F}_p$ -scheme. The *absolute Frobenius* on X is the map $F = (F, F^\#)$

$$\begin{aligned} F &= id_X : X \rightarrow X \\ x &\mapsto x \\ F_U^\# : \mathcal{O}_X(U) &\rightarrow F_*\mathcal{O}_X(U) = \mathcal{O}_X(U) \\ r &\mapsto r^p \end{aligned}$$

where $U \subset X$ is some open set.

It follows from the definition that F is an affine map, so most of the properties and results concerning the Frobenius are affine local in nature. Thus it will not be a great disadvantage to restrict ourselves to the affine case and from now on to only consider commutative rings, at least in the following.

Remark 1.5. Note that for any $\varphi \in \text{Hom}_{\text{Alg}_{\mathbb{F}_p}}(R, S)$ we have $F_S \circ \varphi = \varphi \circ F_R$ by direct computation.

The same holds for F^e , $e \in \mathbb{N}$. i.e. the Frobenius commutes with all ring homomorphisms.

Remark 1.6. For $I \subset R$ an ideal we write $I^{[p]} := RF(I) = R\langle r^p \mid r \in I \rangle$ for the extended ideal. If I is finitely generated it of course suffices to consider the ideal in R generated by the p -th powers of the generators.

We will now discuss the various functors of modules induced by the Frobenius. For this we will quickly recap the general situation for commutative Rings.

Let $f \in \text{Hom}_R(M, N)$ be a morphism of R -modules. In this case f directly induces three different functors on the categories of R -Modules Mod_R .

First of all, there is the *pushforward* or *lower star* functor

$$\begin{aligned} f_* : \text{Mod}_S &\rightarrow \text{Mod}_R \\ M &\mapsto f_* M. \end{aligned}$$

Here $f_* M \cong M \in \text{Ab}$, but the R -module structure is given by $r \cdot m := f(r)m$ for all $r \in R, m \in M$.

This functor has a left adjoint, the *pullback* or *upper star* functor

$$\begin{aligned} f^* : \text{Mod}_R &\rightarrow \text{Mod}_S \\ M &\mapsto f^* M := M \otimes_R f_* S. \end{aligned}$$

The lower star functor has a further right adjoint (note that this does not hold in general for schemes!), called the *upper shriek* functor

$$\begin{aligned} f^! : \text{Mod}_R &\rightarrow \text{Mod}_S \\ M &\mapsto f^! M := \text{Hom}_R(f_* S, M). \end{aligned}$$

So in total we have the following adjunctions:

$$f^* \dashv f_* \dashv f^!$$

where $F \dashv G$ denotes that F is leftadjoint to G .

Exercise. Prove the last statement.

By standard general non-sense arguments it follows that f_* is an exact functor, f^* is right exact and $f^!$ is left exact.

Let us return to the case of the Frobenius. This might get a bit confusing as the Frobenius is an endomorphism of R . We will try to write things in such a way that is not as confusing.

The lower star functor is given by

$$\begin{aligned} F_* : \text{Mod}_R &\rightarrow \text{Mod}_R \\ M &\mapsto F_* M. \end{aligned}$$

We will sometime denote elements of F_*M as $F_*m \in F_*M$. In this notation the r -module structure is given by

$$rF_*(m) = F_*(r^p m).$$

Note that while $F : R \rightarrow R$ is a ring-homomorphism which is not(!) R -linear, $F : R \rightarrow F_*R$ is R -linear. Further notice that the isomorphism $F_*R \cong R \in \text{Mod}_R$ is given by the identity and not by the Frobenius.

Remark 1.7. Note that we have the following:

$$\begin{aligned} R \text{ is reduced} &\Leftrightarrow F : R \hookrightarrow R \\ &\Leftrightarrow F_*R \text{ is a faithful } R \text{ module, i.e. } \text{Ann}_R F_*R = 0 \end{aligned}$$

The upper star functor is given by

$$\begin{aligned} F^* : \text{Mod}_R &\rightarrow \text{Mod}_R \\ M &\mapsto F^*M := M \otimes_R F_*R. \end{aligned}$$

In this case the R -module structure is given by

$$r(m \otimes F_*(a)) = m \otimes F_*(ra),$$

while

$$(rm) \otimes F_*(a) = m \otimes (rF_*(a)) = m \otimes F_*(r^p s).$$

Lastly the upper shriek functor is given by

$$\begin{aligned} F^! : \text{Mod}_R &\rightarrow \text{Mod}_R \\ M &\mapsto F^!M := \text{Hom}_R(F_*R, M). \end{aligned}$$

Here the R -module structure is given by

$$(rf)(F_*a) := f(F_*(ra)).$$

Exercise. Show that F_* is compatible with localisation, i.e. for $S \subseteq R$, $S^{-1}F_*R = F_*(S^{-1}R)$.

Exercise. F is compatible with completion, i.e. the natural map $\widehat{F} : \widehat{R} \rightarrow \widehat{F_*R}$ coincides with the natural map $F_{\widehat{R}} : \widehat{R} \rightarrow F_*\widehat{R}$. Furthermore if R is F -finite, then $\widehat{R} \otimes_R F_*R \cong \widehat{F_*R}$. For an introduction to completions, see [AM18].

To ease the pain of remembering all these definitions we give a commutative diagram for R a domain which helps remembering what happens where:

$$\begin{array}{ccccc} R & \xhookrightarrow{F} & F_*R & & \\ \downarrow = & & \downarrow \cong & & \\ R & \xhookrightarrow{\quad} & R^{1/p} & \xhookrightarrow{\quad} & \overline{Q(R)} \\ \downarrow \cong, F & & \downarrow \cong & & \\ R^p & \xhookrightarrow{\quad} & R & & \end{array}$$

Here the lower two morphisms are induced by the Frobenius.

Example 1.8. Let k be a perfect field and consider $R := k[t_1, \dots, t_n]$. Using the above identifications we obtain the following:

F_*R as an R -module can be identified with R considered as an R^p -module. But

$$R \cong \bigoplus_{0 \leq e_1, \dots, e_n \leq p-1} R^p t_1^{e_1} \dots t_n^{e_n}$$

is a free R^p -Module of rank p^n . Thus F_*R is a flat R -module. This implies further that for this R , F^* is an exact functor.

Definition 1.9. $R \in \text{Alg}_{\mathbb{F}_p}$ is called F -finite, if the Frobenius $F : R \rightarrow R$ is a finite map, i.e. F_*R is a finitely generated R module.

This will be one of the most important notions of this course. The next examples together with example 1.8 will show that a wide variety of rings is in fact F -finite.

Example 1.10. • Assume $R \rightarrow S$ is of finite type, i.e. there is a factorization $R \hookrightarrow R[X_1, \dots, X_n] \rightarrow S$ for some $n \in \mathbb{N}$. Then F -finiteness of R implies that S is F -finite.

- $k \in \text{Field}$. Then: k is F -finite $\Leftrightarrow [k : k^p] < \infty$. This follows from the diagram identifications above.
- $[k : k^p] < \infty \Leftrightarrow k[X_1, \dots, X_n]$ is F -finite for all $n \in \mathbb{N}$. This even holds for $k[[X_1, \dots, X_n]]$.
- If $R \rightarrow T$ is a localization, then R being F -finite implies F -finiteness of T .

However, there are also rings which are not F -finite. An example is the following:

Example 1.11. $\mathbb{F}_p[X_n \mid n \in \mathbb{N}]$ is not F -finite.

The rest of this chapter we will prove (most of) the following theorem:

Theorem 1.12. Let $R \in \text{Alg}_{\mathbb{F}_p}$. If R is F -finite and noetherian, then the following holds:

- a) The regular locus $\text{Reg}(R) := \{\mathfrak{p} \in \text{spec}(R) \mid R_{\mathfrak{p}} \text{ is regular}\}$ is open.
- b) $\dim(R) < \infty$.
- c) $R \cong S/I$ with S regular and $I \subseteq S$ an ideal.
- d) R is excellent.
- e) If R is a domain, the integral closure of R in $Q(R)$ is finite.

Remark 1.13 ([Sta24]). To explain what an excellent ring is, we need some prerequisites. We begin with the geometric property of being a G -ring [Stacks 07GH].

- A k -algebra R for a field k is called *geometrically regular over k* if either one of the following two equivalent properties hold

- a) $k' \otimes_k R$ is regular for every finitely generated field extensions k'/k .
- b) $k' \otimes_k R$ is regular for every finite purely separated field extension k'/k .

- A ring homomorphism $R \rightarrow A$ is called *regular* if it is flat and for every prime $\mathfrak{p} \subset R$ the fibre ring $A \otimes_R \kappa(R_{\mathfrak{p}}) \cong A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is geometrically regular over $\kappa(R_{\mathfrak{p}})$. One can check that this is a local property.
- A ring R is called *G -ring*, if for all primes $\mathfrak{p} \subseteq R$ the map $R_{\mathfrak{p}} \rightarrow R_{\mathfrak{p}}^{\wedge}$ is regular.

The next important property is the property of being $J - 2$ [Stacks07P7].

- A noetherian ring R is called J -0, if the regular locus $\text{Reg}(R) \subseteq \text{spec}(R)$ contains a non-empty open $U \neq \text{Reg}(R)$.
- R is called J -1, if the regular locus $\text{Reg}(R) \subset \text{spec}(R)$ itself is open.
- R is called J -2, if any R -algebra of finite type is J -1.

Notice that a ring can be J -1 without being J -0. However, this cannot happen for noetherian domains or more generally for nonzero reduced noetherian rings.

The last important property is being universally catenary [Stacks00NH].

- A ring R is called *catenary* if for every pair of primes $\mathfrak{p} \subseteq \mathfrak{q} \in \text{spec}(R)$ there exists an integer bound on the length of ascending primes $\mathfrak{p} = \mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \dots \subseteq \mathfrak{p}_n = \mathfrak{q}$ and all maximal chains have the same length.
- A ring is called *universally catenary* if every R -algebra of finite type is catenary.

Now we can state the definition of excellent [Stacks07QS].

- A ring R is called *quasi-excellent* if R is noetherian, a G -ring and a J -2 ring.
- R is called *excellent* if R is quasi-excellent and universal catenary

The properties described in theorem (1.12) are in some sense a generalization of properties enjoyed by finite type k -algebras for a field k :

Exercise. Let k be a field and R be a k -algebra of finite type. Proof the following:

- $\dim(R) < \infty$
- The regular locus of R is open
- We have $k \hookrightarrow k[X_1, \dots, X_n] \twoheadrightarrow R$ where the first map is smooth and $k[X_1, \dots, X_n]$ is regular.
- R is excellent.
- R has a dualizing complex.
- For R an integral domain, the integral closure of R in its fraction field $Q(R)$ is finite over R .
- We have $R \hookrightarrow R[X_1, \dots, X_n] \twoheadrightarrow S$ for some ? R -algebra S

After this glimpse of what we want to show, let us begin by reminding ourselves how a regular ring was defined

Definition 1.14. A noetherian local ring $(R, \mathfrak{m})$ is called *regular* if any of the following equivalent definitions hold:

- 1) $\dim(R) = \dim_k \mathfrak{m}/\mathfrak{m}^2$ for $k = R/\mathfrak{m}$ the residue field.
- 2) Every finitely generated R -module M the projective dimension $\text{pd}_R M$ is finite.
- 3) $\text{pd}_R R/\mathfrak{m} < \infty$

The last two equivalences use the theorem of Auslander-Buchsbaum-Serre, see [Mat89, Ch. 19].

Definition 1.15. A noetherian ring R is called *regular*, if the local ring $R_{\mathfrak{p}}$ is regular for all $\mathfrak{p} \in \operatorname{spec}(R)$.

This global definition is compatible with the local definition:

Proposition 1.16. *If $(R, \mathfrak{m})$ is a noetherian local regular ring, then $R_{\mathfrak{p}}$ is regular for all $\mathfrak{p} \in \operatorname{spec}(R)$.*

The proof is quite easy using the homological definition of regularity and left as an exercise to the reader.

The next proposition summarizes important properties of regular rings.

Proposition 1.17. *Let R be a ring.*

- *Regularity of R implies that R is normal.*
- *A local noetherian regular ring $(R, \mathfrak{m})$ is a UFD, see [Mat89, Ch. 20] or [Wei94, Proposition 4.4.5].*
- *(Cohen structure theorem) Assume that $(R, \mathfrak{m})$ is a local, complete regular ring and that $k \subseteq R$ for some field k . Then*

$$R \cong k[[X_1, \dots, X_n]] \cong k[X_1, \dots, X_n]^{\wedge}$$

- *More generally, if R is a complete local noetherian ring then there exists a regular complete ring S together with a surjection $S \twoheadrightarrow R$.*

Regular rings are something like „well behaved rings with properties that one would wish for “. They also appear in Algebraic geometry. There a variety is smooth if and only if its ring of functions is regular. So for geometry, the property of a ring to be regular is quite important. Now, a nice thing of positive characteristic geometry is, that the Frobenius is in fact able to detect regularity:

Theorem 1.18 (Kunz). *Let $R \in \operatorname{Alg}_{\mathbb{F}_p}$ be noetherian. Then the following statements are equivalent*

- *R is regular,*
- *F_*R is a flat R module,*
- *$F : R \rightarrow R$ is flat.*

Corollary 1.19. *Let $R \in \operatorname{Alg}_{\mathbb{F}_p}$ be noetherian and F -finite. Then the regular locus is open.*

Proof. R being F -finite implies that F_*R is a finitely generated R -Module by Definition. It is a fact that this implies that the set of all primes $\mathfrak{p} \in \operatorname{spec}(R)$ such that $(F_*R)_{\mathfrak{p}} = F_*(R_{\mathfrak{p}})$ is free is open. This proves the claim using that in this setting projective equals free equals flat and that by Kunz theorem the stalks $(F_*R)_{\mathfrak{p}}$ are flat. $\square$

Exercise. Let X be a noetherian scheme and $\mathcal{F}$ be a coherent sheaf. If the stalk $\mathcal{F}_x$ at $x \in X$ is a free $\mathcal{O}_X$ -module, then there exists a neighborhood U of x such that $\mathcal{F}|_U$ is free. See [Har13, II, Ex. 5.7].

Remark 1.20. In classical algebraic geometry there is also a module which can be used to measure regularity, or to be more precise, smoothness, which is closely related to regularity in this context.

If R is a finitely generated k algebra for k an algebraically closed field, then the module of Kähler differentials $\Omega_{R/k}^1$ measures smoothness. That is, by the Jacobi criterion the finitely generated R -module $\Omega_{R/k}^1$ is projective if and only if R/k is smooth.

We will now try to proof Kunz theorem. The first step is to reduce the theorem greatly.

Lemma 1.21. *For the proof of Kunz theorem we can reduce to the case of $(R, \mathfrak{m})$ local and complete.*

Proof. First of all regularity (by definition) and flatness (see [AM18, Proposition 3.10]) are local in nature, so we can reduce to the local case.

Further $(R, \mathfrak{m})$ is regular $\Leftrightarrow (\widehat{R}, \widehat{\mathfrak{m}})$ is regular. This follows from

$$\dim \widehat{R} = \dim R = \dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim_k \widehat{\mathfrak{m}}/\widehat{\mathfrak{m}}^2.$$

Lastly $F_* R$ is flat over $R \Leftrightarrow (F_* R)^\wedge = F_*(R^\wedge)$ is flat over $R^\wedge$. Necessity follows from flatness being stable under basechange. For the sufficient part see [Stacks 039V]. $\square$

We are now able to proof the easy direction of Kunz theorem.

Proof of R regular implies $F : R \rightarrow R$ is flat. We have the following chain of maps for a noetherian complete local regular ring: (**redo in tikzcd**)

$$\begin{aligned} R &\cong k[[X_1, \dots, X_n]] \\ &\hookrightarrow k[[X_1^{1/p}, \dots, X_n^{1/p}]] \\ &\hookrightarrow k^{1/p} \otimes_k k[[X_1^{1/p}, \dots, X_n^{1/p}]] \\ &\rightarrow k^{1/p}[[X_1^{1/p}, \dots, X_n^{1/p}]] \\ &\cong R^{1/p} \end{aligned}$$

here the first isomorphism is due to Cohen's structure theorem, the next two inclusions are free maps, the next map is induced by completion and an isomorphism if $[k^{1/p} : k] < \infty$. It is not hard to see that this composition is equal to the Frobenius. $\square$

For the other direction we need a lot of homological algebra and the theory of perfect rings. For the homological algebra part we refer to [Wei94, Ch. 10]. The most important part is the definition of *derived functors* and the *derived category*.

Definition 1.22. $R \in \text{Alg}_{\mathbb{F}_p}$ is called *semi-perfect* if the Frobenius endomorphism is surjective. R is called *perfect* if the Frobenius is an isomorphism.

Note that perfect is equivalent to semi-perfect and reduced.

Example 1.23. The following two examples are common examples of perfect rings

- Perfect fields are perfect (Hooray)
- $\mathbb{F}_p[t]_{\text{perf}} := \mathbb{F}_p[t, t^{1/p}, t^{1/p^2}, \dots]$ is perfect.

Exercise. A noetherian perfect $\mathbb{F}_p$ -algebra R is zero dimensional and reduced, i.e. R is a product of perfect fields.

There are two canonical ways to form a perfect $\text{Alg}_{\mathbb{F}_p}$ out of a given $\text{Alg}_{\mathbb{F}_p}$.

Definition 1.24. Let $R \in \text{Alg}_{\mathbb{F}_p}$.

- The *perfection* of R is the ring R_{perf} defined as the colimit over the Frobenius diagram, i.e.

$$R_{\text{perf}} := R^{1/p^\infty} := R^\infty := \text{colim}_F(R) := \text{colim} \left(R \xrightarrow{F} R \xrightarrow{F} \dots \right).$$

- The *tilt* of R is the ring $R^\flat$ defined as the limit of the Frobenius diagram:

$$R^\flat := \lim_F (R) = \lim \left(\dots \xrightarrow{F} R \xrightarrow{F} R \right)$$

As one checks readily these two definitions in fact define perfect algebras.

Remark 1.25. Let $R \in \text{Alg}_{\mathbb{F}_p}$.

- 1 In the special case that R is an integral domain the perfection can be rewritten as

$$R_{\text{perf}} = \text{colim} \left(R \hookrightarrow R^{1/p} \hookrightarrow R^{1/p^2} \hookrightarrow \dots \right) = \bigcup_{e \in \mathbb{N}} R^{1/p^e}.$$

So in fact the perfection is „nothing else“ then adjoining all p -th power roots to a ring.

- 2 the canonical map of the 0-th component $R \rightarrow R_{\text{perf}}$ and the canonical projection to the 0-th component $R^\flat \rightarrow R$ are left, respectively right adjoint to the forgetful functor

$$U : \text{Alg}_{\mathbb{F}_p}^{\text{perf}} \rightarrow \text{Alg}_{\mathbb{F}_p}.$$

where $\text{Alg}_{\mathbb{F}_p}^{\text{perf}}$ is the full subcategory of $\text{Alg}_{\mathbb{F}_p}$ consisting of perfect $\mathbb{F}_p$ -Algebras.

- 3) The map $R \rightarrow R_{\text{perf}}$ induces a homeomorphism on spectra:

$$\text{spec}(R_{\text{perf}}) \xrightarrow{\cong} \text{spec}(R) \in \text{Top}.$$

- 4) If R is noetherian, then $\text{spec}(R_{\text{perf}})$ is a noetherian topological space, even though R_{perf} itself is in general not noetherian.

Our goal is still to prove the other direction of the theorem of Kunz. We want to give an elegant proof, but for this we need the following important result of Bhargav Bhatt and Peter Scholze from the paper [BS15].

Theorem 1.26. *Let $S \twoheadrightarrow R$ be a surjection of rings with S regular and R noetherian. Then the homological, or global, dimension of R_{perf} is finite, i.e. there exists $n \in \mathbb{N}$ such that for every R_{perf} -module M , we have $\text{pd}(M) \leq n$.*

So even though R_{perf} is in general not noetherian, we could call R_{perf} somewhat „regular“. The theorem is a corollary of the following, known as the „Tor-Vanishing“ theorem.

Proposition 1.27. *Let $R \leftarrow S \rightarrow T$ be a diagram of perfect rings. Then $\text{Tor}_i^S(R, T) = 0$ for all $i > 0$. Equivalently $R \otimes_S^{\mathbb{L}} T$ is isomorphic to $R \otimes_S T$ concentrated in degree 0 in the derived category of R -modules.*

Proof of Thm.1.26. Before we start the proof here are a few important facts:

- Note that then $R_{\text{perf}} \cong S_{\text{perf}}/I_{\text{perf}}$ as colimits commute. Here $I_{\text{perf}} := \text{colim}_F (I)$.
- As R_{perf} is reduced, I_{perf} is a radical ideal of S_{perf} .
- We have $\text{Ext}_{R_{\text{perf}}}^i(M, N) := H_i \left(\text{Hom}_{R_{\text{perf}}}(P_\bullet, N) \right) = 0$ for all $i \geq n$ if and only if $\text{pd}(M) = n$, see [Wei94, pd lemma 4.1.6]. So to show that $\text{gldim}(R_{\text{perf}}) = n$ is equivalent to show that for all $M, N \in R\text{-Mod}$ we have $\mathbb{R}\text{Hom}_{R_{\text{perf}}}(M, N) \in \mathcal{D}^{\leq n}(R_{\text{perf}})$.

Now we are ready for the proof, which basically consists of two clever calculations.

$$\begin{aligned}
M &\cong M \otimes_{R_{\text{perf}}}^{\mathbb{L}} R_{\text{perf}} \cong M \otimes_{R_{\text{perf}}}^{\mathbb{L}} (R_{\text{perf}} \otimes_{S_{\text{perf}}} R_{\text{perf}}) \\
&\cong M \otimes_{R_{\text{perf}}}^{\mathbb{L}} (R_{\text{perf}} \otimes_{S_{\text{perf}}}^{\mathbb{L}} R_{\text{perf}}) \\
&\cong (M \otimes_{R_{\text{perf}}}^{\mathbb{L}} R_{\text{perf}}) \otimes_{S_{\text{perf}}}^{\mathbb{L}} R_{\text{perf}} \\
&\cong M \otimes_{S_{\text{perf}}}^{\mathbb{L}} R_{\text{perf}}
\end{aligned}$$

Here we used the Tor-vanishing in the third isomorphism. With this we get the following magical result:

$$\begin{aligned}
\mathbb{R}\text{Hom}_{R_{\text{perf}}}(M, N) &\cong \mathbb{R}\text{Hom}_{R_{\text{perf}}}(M \otimes_{S_{\text{perf}}}^{\mathbb{L}} R_{\text{perf}}, N) \\
&\cong \mathbb{R}\text{Hom}_{S_{\text{perf}}}(M, \mathbb{R}\text{Hom}_{R_{\text{perf}}}(R_{\text{perf}}, N)) \\
&\cong \mathbb{R}\text{Hom}_{S_{\text{perf}}}(M, N).
\end{aligned}$$

So the homomorphisms of any two modules M and N of a perfected quotient ring are the same as the homomorphisms of M and N regarded as modules over the ring which was quotient. In our hypothesis S was a regular ring, so S_{perf} too is regular. Hence we can assume without loss of generality that R itself was regular to begin with.

R regular implies that the localization on every prime ideal is a UFD and hence R is reduced. But then using Remark 1.25 part 1. we see that

$$R_{\text{perf}} = \text{colim} \left(R \hookrightarrow R^{1/p} \hookrightarrow R^{1/p^2} \hookrightarrow \dots \right) = \bigcup_{e \in \mathbb{N}} R^{1/p^e}.$$

Thus we have in fact natural inclusions $R^{1/p^e} \hookrightarrow R_{\text{perf}}$, giving every R_{perf} -Module M a R^{1/p^e} -module structure. We define a family of modules as

$$M_e := M \otimes_{R^{1/p^e}} R_{\text{perf}},$$

together with transition maps

$$\begin{aligned}
\varphi : M_e &= M \otimes_{R^{1/p^e}} R_{\text{perf}} \rightarrow M \otimes_{R^{1/p^{e+1}}} R_{\text{perf}} = M_{e+1} \\
m \otimes r &\mapsto m \otimes r.
\end{aligned}$$

As the reader may check readily, we have $\text{colim} \left(\dots \rightarrow M_e \xrightarrow{\varphi} M_{e+1} \xrightarrow{\varphi} \dots \right) \cong M \otimes_{R_{\text{perf}}} R_{\text{perf}} \cong M$ as an R_{perf} -module.

Now we are basically done. For this, note that we can resolve $M \cong \text{colim}_{\varphi} (M_e)$ as the following short exact sequence

$$0 \longrightarrow \bigoplus M_e \longrightarrow \bigoplus M_e \longrightarrow \text{colim}_{\varphi} (M_e) \longrightarrow 0$$

Using the long exact $\text{Ext}_{R_{\text{perf}}}$ -sequence, we see that

$$\text{pd}_{R_{\text{perf}}}(M) \leq \text{pd}_{R_{\text{perf}}}(\bigoplus M_e) + 1 \leq \sup_e \left(\text{pd}_{R_{\text{perf}}}(M_e) \right) + 1.$$

But $R^{1/p^e} \cong R \in \text{Alg}_{\mathbb{F}_p}$, so we get that

$$\sup_e \left(\text{pd}_{R_{\text{perf}}}(M_e) \right) + 1 = \sup_e \left(\text{pd}_{R_{\text{perf}}}(M \otimes_R R_{\text{perf}}) \right) + 1.$$

However every projective resolution $P_{\bullet} \rightarrow M$ induces a projective resolution $P_{\bullet} \otimes_R R_{\text{perf}} \rightarrow M \otimes_R R_{\text{perf}}$, using that being projective is preserved under base change and R_{perf} is flat as an R module. So

$$\text{pd}_{R_{\text{perf}}}(M \otimes_R R_{\text{perf}}) \leq \text{pd}_R(M) \leq \dim(R) < \infty$$

proving the claim. $\square$

With this the direction of Kunz theorem yet to prove is nothing more then a corollary.

Corollary 1.28 (Kunz Part II). *Let R be a noetherian $\mathbb{F}_p$ -Algebra. If the Frobenius $F : R \rightarrow R$ is flat, then R is regular.*

Proof. By lemma 1.21 we can assume that R is a complete noetherian local ring. Assume that the Frobenius is flat.

By the Cohen-structure theorem we can write $R \cong S/I$ for $S = k[[X_1, \dots, X_n]]$ a regular power series ring. R_{perf} is flat over R as a colimit of flat R -modules. In fact, as the map on spectra is a homeomorphism, R_{perf} is even faithfully flat. It follows for all $i > \dim(R_{\text{perf}}) + 1 = \dim(R) + 1$

$$\begin{aligned} \text{Tor}_i^R(M, N) = 0 &\Leftrightarrow R_{\text{perf}} \otimes_R \text{Tor}_i^R(M, N) = 0 \\ &\Leftrightarrow \text{Tor}_i^{R_{\text{perf}}}(R_{\text{perf}} \otimes_R M, R_{\text{perf}} \otimes_R N) = 0 \end{aligned}$$

But by the above proposition, the last statement is true, so R has finite global dimension. This is by the Auslander-Buchsbaum-Serre theorem equivalent to R being regular, finishing the proof. $\square$

In fact, the proof shows even a bit more.

Corollary 1.29. *Let R be noetherian. Then the following statements are equivalent.*

- 1) R is regular.
- 2) $R \rightarrow R_{\text{perf}}$ is faithfully flat.
- 3) There exist a faithfully flat extension $R \rightarrow S$ with S perfect.

Here the last equivalence is an exercise for the reader. This formulation holds even in greater generality

Theorem 1.30 (Ma, Jenga, Bhatt). *Let R be a noetherian local ring of mixed characteristics. then the following statements are equivalent.*

- 1) R is regular.
- 2) There exist a faithfully flat extension $R \rightarrow S$ with S perfectoid.
- 3) The map $R \rightarrow R^+$ is flat, where R^+ is the absolute integral closure of R .

This theorem however is way over the scope of this course, so we will not prove this here.

2 F -splitting

In this chapter we will consider a weaker notion of regularity which can again be measured by the Frobenius.

Proposition 2.1. *If $R \in \text{Alg}_{\mathbb{F}_p}$ is F -finite and regular then there exists a split $\varphi : F_*R \rightarrow R$ of the Frobenius as a R -module homomorphism, i.e. $\varphi \circ F = \text{id}_R$.*

Proof. Note that a splitting of $F : R \rightarrow F_*R \in \text{Mod}_R$ is the same datum as that the composition

$$\text{Hom}_R(F_*R, R) \xrightarrow{\circ F} \text{Hom}_R(R, R) \cong R$$

given by $\varphi \mapsto \varphi \circ F \mapsto \varphi(F(1)) = \varphi(F_*1)$, which is just the evaluation at F_*1 , is surjective as a R -module homomorphism. But surjectivity can be checked locally on every maximal ideal $\mathfrak{m}$. Hence it suffices to show the proposition for R regular, local and F finite. But then, by Kunz theorem, F is flat and hence F_*R is a finitely generated flat R -module. But in our situation, finitely generated and flat is the same as finitely generated and projective is the same as locally free is the same as free. But then just choose a map

$$\begin{aligned} \varphi : F_*R &\cong \bigoplus_{i=1}^d R \twoheadrightarrow R, \\ F_*1 &\mapsto a \mapsto 1. \end{aligned}$$

This φ splits F . □

That regular together with F -finiteness implies this weaker condition of a Frobenius splitting leads to the following

Definition 2.2. $R \in \text{Alg}_{\mathbb{F}_p}$ F -finite is called F -split if and only if there exists a split $\varphi : F_*R \rightarrow R$ of the Frobenius, if and only if the evaluation map $\text{ev}_{F_*1} : \text{Hom}_R(F_*R, R)$ at F_*1 is surjective.

The proposition above shows exactly that an R regular, F -finite ring is F -split.

Exercise. For an F -finite ring R the following are equivalent.

- 1) R is F -split.
- 2) F^e is split for some $e > 0$.
- 3) F^e is split for any $e > 0$.
- 4) There exists a surjection $F_*^e R \rightarrow R$ for some $e > 0$.
- 5) There exists a surjection $F_*^e R \rightarrow R$ for any $e > 0$.

Theorem 2.3. *Let $R \hookrightarrow S \in \text{Alg}_{\mathbb{F}_p}$ be a ring extension together with a surjection $\varphi : S \twoheadrightarrow R \in R\text{-Mod}$. Then, if S is F -split, R is.*

Proof. A map f is a splitting if and only if $1 \in \text{Im}(f)$, i.e there exists an element $a \in S$ with $\varphi(a) = 1$. Thus we can assume that φ is a split of $R \hookrightarrow S$ by modifying φ as

$$\begin{aligned} S &\xrightarrow{a} S \xrightarrow{\varphi} R \\ 1 &\mapsto a \mapsto 1 \end{aligned}$$

In this case, using that S is F -split, we have the following diagram where the upper horizontal composition is the identity and the right vertical arrows define a splitting:

$$\begin{array}{ccccc} S & \longrightarrow & F_*S & \twoheadrightarrow & S \\ \uparrow & & \uparrow & & \uparrow \downarrow \\ R & \longrightarrow & F_*R & & R \end{array}$$

Using the split we can fill the remaining arrow giving a retract diagram. $\square$

This theorem gives us a wide range of F -split rings.

Example 2.4. Consider k a field of characteristic p and $R := k[X^2, XY, Y^2]$. Then the ring inclusion

$$R \hookrightarrow k[X, Y] \quad (2.0.1)$$

is split as one sees writing R and $k[X, Y]$ as modules, so R is F split.

More generally

Corollary 2.5. *Every F -finite noetherian ring which is direct summand of a regular ring is F -split.*

Example 2.6. • Every Veronese subring $R_d = \bigoplus_{m \in \mathbb{N}} k[X_1, \dots, X_n]_{md} \subseteq k[X_1, \dots, X_n]$ is F -split.

- Let G be a finite group acting k -linear on $k[X_1, \dots, X_n]$ and assume that $|G| \nmid p = \text{char}(k)$. The $k[X_1, \dots, X_n]^G \hookrightarrow k[X_1, \dots, X_n]$ is split with the split given by the Reynolds operator $R(f) := \frac{1}{|G|} \sum_{g \in G} gf$. Thus, the ring of invariants is F -split.

From now on we will focus on the following question: For $S \in \text{Alg}_{\mathbb{F}_p}$ regular, under which conditions on an ideal $I \subset S$ is R/I F -split?

For this consider the following

Example 2.7. Let $S = k[X_1, \dots, X_n]$. In this case, $F_*^e S$ is a free S -module with basis

$$\{F_* a_j X^\lambda \mid 0 \leq \lambda_i \leq p^e - 1 \text{ and } a_j \in B\} \quad (2.0.2)$$

as $k[X_1, \dots, X_n] \hookrightarrow k^{1/p^e}[X_1, \dots, X_n] \hookrightarrow k^{1/p^e}[X_1^{1/p^e}, \dots, X_n^{1/p^e}] = S^{1/p^e}$.

Explicitly, for $n = 0$ one checks readily that $\text{Hom}_k(F_*^e k, k) = (F^e)^! k$ is a free $F_*^e k$ vector space of dimension 1. Note that $(F^e)^!$ carries a k - $F_*^e k$ -bimodule structure.

More generally, using the example to reduce to k perfect,

Proposition 2.8. *Let k be a perfect field of characteristic p and $S = k[X_1, \dots, X_n]$. Then $\text{Hom}_S(F_*^e S, S) \cong F_*^e$ as F_*^e modules. The isomorphism is on generators given by sending $F_*^e X^\lambda$ to $\Phi_S(F_*^e X^\lambda) = 1$ if $\lambda = (p^e - 1, \dots, p^e - 1)$ and 0 else.*

Proof. We know that the projections generate $\text{Hom}_S(F_*^e S, S)$ as an S -module. So all we have to show is that these projections are multiples of Φ_S which is an easy calculation. $\square$

From now on we consider the more general case $S \twoheadrightarrow R/I$ where S is regular and $I = (f_1, \dots, f_n)$ is of finite presentation. In light of the above proposition one can ask if and how $\text{Hom}_S(F_*^e S, S)$ and $\text{Hom}_R(F_*^e R, R)$ are connected.

We will need the following facts and definition for the rest of this chapter

- $\text{Hom}_S(F_* S, S)$ is an S - $F_* S$ -bimodule.
- $I^{[p^e]} := (f_1^{p^e}, \dots, f_n^{p^e})$.

- The colon ideal is defined as $I :_S J := \{x \in S \mid xJ \subset I\}$.

Exercise. Let $\varphi \in \text{Hom}_S(F_*S, S)$ and $u \in I^{[p]} :_S J$. Then $\psi := \varphi \circ F_*u : F_*S \rightarrow S$ satisfies $\psi(F_*J) \subseteq I$.

In general, if $\varphi \in \text{Hom}_S(F_*S, S)$ such that $\varphi(F_*J) \subseteq I$, then φ descends uniquely to a morphism $\bar{\varphi} : \text{Hom}_R(F_*R, R)$:

$$\begin{array}{ccccc} F_*I & \hookrightarrow & F_*S & \xrightarrow{\varphi} & S & \longleftarrow & I \\ & & \downarrow & & \downarrow & & \\ & & F_*R & \xrightarrow{\bar{\varphi}} & R & & \end{array}$$

We have the following diagram:

$$\begin{array}{ccc} \text{Hom}_S(F_*S, S) \cdot F_*(I^{[p]} :_S J) & \hookrightarrow & \{\varphi \in \text{Hom}_S(F_*S, S) \mid \varphi(F_*I) \subseteq I\} \hookrightarrow \text{Hom}_S(F_*S, S) \\ & \searrow & \downarrow \\ & & \text{Hom}_R(F_*R, R) \end{array}$$

In the next lemma we will show that for S regular, the first horizontal inclusion is an equality and the diagonal map is surjective.

Lemma 2.9. *Let $S \in \text{Alg}_{\mathbb{F}_p}$ be regular. Then*

- i) $\varphi(F_*J) \subset I$ for all $\varphi \in \text{Hom}_S(F_*S, S)$ if and only if $J \subset I^{[p]}$.
- ii) $\text{Hom}_S(F_*S, S) \cdot F_*(I^{[p]} :_S J) = \{\varphi \in \text{Hom}_S(F_*S, S) \mid \varphi(F_*I) \subseteq I\}$.

Proof. We will only proof the lemma for $S = k[X_1, \dots, X_n]$ with k a perfect field (in fact, the lemma can always be reduced to this case). Under this assumption, F_*S is a free module with basis $\{F_*g_i\}_{i \in E}$. Thus $F_*S \cong \bigoplus_{i \in E} S \cdot F_*g_i \cong \bigoplus_{i \in E} S$ as S -modules. Consider the projections

$$\pi_i : F_*S \rightarrow S \cdot F_*g_i \cong S.$$

We also have

$$F_*I^{[p]} = I \cdot F_*S = I \cdot \bigoplus_{i \in E} S \cdot F_*g_i \cong \bigoplus_{i \in E} I \cdot F_*g_i.$$

Now assume $\varphi(F_*J) \subset I$ for all $\varphi \in \text{Hom}_S(F_*S, S)$. In particular $\pi_i(F_*J) \subseteq I$ for all $i \in E$. Combing these thoughts we get

$$F_*J \subseteq \bigoplus_{i \in E} I \cdot F_*g_i \cong F_*I^{[p]}.$$

Thus, on the level of underlying abelian groups

$$J \cong F_*J \subseteq F_*I^{[p]} \cong I^{[p]}.$$

This shows $\Rightarrow$ in i). The other direction is clear.

Regarding ii), we've already seen $\subseteq$, so we only have to show the inclusion $\supseteq$. For this let $\psi \in \text{Hom}_S(F_*S, S)$ with $\psi(F_*I) \subseteq I$. Let Φ_S be the generator of $\text{Hom}_S(F_*S, S)$ as an F_*S, S -module, i.e. there exists an $u \in S$ such that $\psi = \Phi_S F_*u$. It follows that

$$\psi(F_*J) = \Phi_S(F_*u \cdot J) \subseteq I.$$

Using that Φ_S is a generator and i) we conclude $uJ \subset I^{[p]}$ and hence $u \in I^{[p]} :_S J$ which proofs the claim. $\square$

The next theorem gives us the answer to our question how the different morphism sets are connected.

Theorem 2.10 (Fedder). *Let $S \in \text{Alg}_{\mathbb{F}_p}$ be a regular, F -finite ring and $R \cong S/I$ be a quotient ring as above. Then the obvious map*

$$\text{Hom}_S(F_*S, S) \cdot F_*(I^{[p]} :_S I) \twoheadrightarrow \text{Hom}_R(F_*R, R)$$

*is surjective and the kernel is given by the set $\text{Hom}_S(F_*S, S) \cdot F_*I^{[p]}$. Equivalently*

$$\text{Hom}_R(F_*R, R) \cong \text{Hom}_S(F_*S, S) \otimes_{F_*S} \frac{F_*(I^{[p]} :_S I)}{F_*I^{[p]}}.$$

Proof. Take some $\bar{\alpha} \in \text{Hom}_R(F_*R, R)$. We have the following diagram with exact rows

$$\begin{array}{ccccc} F_*I & \hookrightarrow & F_*S & \twoheadrightarrow & F_*R \\ & & & & \downarrow \bar{\alpha} \\ I & \hookrightarrow & S & \twoheadrightarrow & R \end{array}$$

Notice that by Kunz theorem and F -finiteness F_*S is a finite flat S -module, hence projective. So using the lifting property of projective modules we see that there exists an $\alpha : F_*S \rightarrow S$ filling the diagram. Using the commutativity and the exactness we see $\alpha(F_*I) \subset I$. It follows from Lemma 2.9 ii) with $J = I$ that $\alpha \in \text{Hom}_S(F_*S, S) \cdot F_*(I^{[p]} :_S I)$, showing surjectivity.

Let us now discuss the kernel. Let $\varphi \in \text{Hom}_S(F_*S, S)$ be such, that $\varphi(F_*I) \subset I$. Let $\bar{\varphi} : F_*R \rightarrow R$ be the induced morphism. The $\bar{\varphi} = 0$ means exactly that $\bar{\varphi}(F_*R) = 0$ or equivalently, that $\varphi(F_*S) \subseteq I$. But this implies $\varphi \in \text{Hom}_S(F_*S, S) \cdot F_*I^{[p]}$, by Lemma 2.9 ii) with $J = S$. Note for this that $I^{[p]} :_S S = I^{[p]}$ as $1 \in S$. This shows the inclusion $\text{Ker} \hookrightarrow \text{Hom}_S(F_*S, S) \cdot F_*I^{[p]}$. The other inclusion is trivial. $\square$

This theorem has the following interesting implication regarding the F -split locus of $R = S/I$:

Theorem 2.11. *Let $R = S/I$ with $S \in \text{Alg}_{\mathbb{F}_p}$ regular. Then*

i) $U := \{\mathfrak{p} \in \text{spec}(R) \mid R_{\mathfrak{p}} \text{ is } F\text{-split}\}$ is open in $\text{spec}(R)$.

*ii) $J_e := \text{Im} \left(\text{Hom}_S(F_*S, S) \cdot F_*(I^{[p]} :_S I) \xrightarrow{\text{ev}_{F_*1}} S \right)$ is equal to the set $= \Phi^e(F_*(I^{[p]} :_S I)) \subseteq S$. Here, $\Phi^e = \Phi \circ F_*\Phi^{e-1}$ is the e -fold composition of the generator of $\text{Hom}_S(F_*S, S)$ as an F_*S -module. Furthermore, $V(J_e) = \text{spec}(R) \setminus U$ is the F -nonsplit locus, independent of e .*

Proof. As we already discussed multiple times, R is F -split if and only if the evaluation ev_{F_*1} is surjective. Set $J' := \text{Im}(\text{ev}_{F_*1}) \subseteq R$. Then R is F -split if and only if $R/J' = 0$. This implies that $R_{\mathfrak{p}}$ is F -split if and only if $R_{\mathfrak{p}}/J'_{\mathfrak{p}} = 0$. In particular, U is the complement of the carrier of the finitely generated R -module R/J' and hence is open.

For part ii) notice that if $\pi : S \twoheadrightarrow R$ then $J = \pi^{-1}(J')$. Furthermore

$$\begin{array}{ccc} \text{Hom}_S(F_*^e S) \cdot F_*^e(I^{[p]} :_S I) & \twoheadrightarrow & \text{Hom}_R(F_*^e R, R) \\ \left(\begin{array}{ccc} \downarrow \text{ev}_{F_*^e 1} & & \text{ev}_{F_*^e 1} \downarrow \\ S & \xrightarrow{\pi} & R \\ \uparrow & & \uparrow \\ J_e = \pi^{-1}(J'_e) & \twoheadrightarrow & J'_e \end{array} \right) \end{array}$$

As $F : R \rightarrow F_*R$ is split if and only if $F^e : R \rightarrow F^e : R \rightarrow F_*^e R$ is split for some e we have that $V(J_e) = V(J_f)$ for all e, f and hence

$$\begin{array}{ccc} \operatorname{spec}(R) & \hookrightarrow & \operatorname{spec}(S) \\ \uparrow & & \uparrow \\ V(J'_e) & \xrightarrow{=} & V(J_e) \end{array}$$

proving the claim. $\square$

Remark 2.12. The ideals $J_e \supseteq J_{e+1} \supseteq \dots$ form a descending chain terminates if R is Gorenstein, so in particular if R is a complete intersection and thus, as a special case, if $R = S/(f)$ for some non-zero divisor $f \in R$. In general it is not known whether this claim holds in general.

In simple cases, as in the last one of the previous remark, we can actually calculate J_e :

Example 2.13. Let $R = S/(f)$ as in the remark, $S = k[X_1, \dots, X_n]$ with k perfect. Then one readily checks $(f^{[p]} :_S f) = (f^{p^e-1})$ and $J_e = \Phi^e(F_*(f^{p^e-1}))$. But we can expand f^{p^e-1} in the basis of $\operatorname{Hom}_S(F_*^e S, S)$ as an $F_*^e S$ module as

$$f^{p^e-1} = \sum_{0 \leq \lambda < p^e} f_\lambda F_*^e(X^\lambda).$$

So $J_e = (f_\lambda | 0 \leq \lambda < p^e)$.

Another consequence is the following criterion for a quotient of a regular ring at some point to be F -split.

Theorem 2.14 (Fedder Criterion). *Let $S \in \operatorname{Alg}_{\mathbb{F}_p}$ be F -finite and regular, $R = S/I$. Then R is F -split at $\mathfrak{p} \in \operatorname{spec}(R)$ if and only if $(I^{[e]} :_S I) \subseteq \mathfrak{p}^{[e]}$.*

Proof. By localizing at $\mathfrak{p}$ we can assume without loss of generality that $(R, \mathfrak{m})$ is local. Then R is F -split if and only if the upper horizontal arrow in the following diagram is surjective

$$\begin{array}{ccc} \operatorname{Hom}_R(F_*R, R) & \xrightarrow{\operatorname{ev}_{F_*1}} & R \\ \uparrow & & \uparrow \\ \operatorname{Hom}_S(F_*S, S) \cdot F_*(I^{[e]} :_S I) & \longrightarrow & S \end{array}$$

This is the case if and only if $J_1 = \Phi_S F_*(I^{[e]} :_S I) = S$ which is equivalent to $(I^{[e]} :_S I) \not\subseteq \mathfrak{m}^{[p]}$. For the second to last equality note that if the lower horizontal map in the diagram is not surjective, then the image of the map is contained in $\mathfrak{m}_S$ hence the composition is not surjective and the upper horizontal map cannot possibly be surjective. For the last equivalence use Lemma 2.9 i) with $I = \mathfrak{m}$ and $J = (I^{[e]} :_S I)$. $\square$

Using this criterion we are now able to identify plenty of F -split rings.

Example 2.15. Let $S := k[X, Y, Z]_{(X, Y, Z)} \ni f$.

- Take $f = XYZ$. Then $(f^p :_S f) = (f^{p-1}) = (X^{p-1}Y^{p-1}Z^{p-1}) \not\subseteq (X^p, Y^p, Z^p)$. So S/xyz is F -split.
- Take $f = XY - Z^2$. Then $F^{p-1} = X^{p-1}Y^{p-1} + \text{terms with } Z \notin (X^p, Y^p, Z^p)$. Hence $S/(XY - Z^2)$ is F -split. So the standard cone in $\mathbb{A}_{\mathbb{F}_p}$ is F -split.
- Assume $\operatorname{char} k = 2$. Then $p - 1 = 1$ and hence R is F -split if and only if $f \notin (X^2, Y^2, Z^2)$.

- Take $f = X^3 + Y^3 + Z^3$. To see when this elliptic curve is F -split, assume $p \geq 5$, the other two cases are easily done by hand. Then

$$f^{p-1} = \text{Terms with powers higher than } p\alpha X^{p-1}Y^{p-1}Z^{p-1}$$

and α is 1 if and only if p is congruent 1 modulo 3, and 0 else. So the elliptic curve $S/(f)$ is F -split if and only if $p \equiv 1 \pmod{3}$ and $p \geq 5$.

Before we proceed to the next chapter, let us remark that the theory of F -split Rings can be generalized to so called F -pure rings. For more information on this topic we refer the interested reader to [MP21], chapter 2.

3 F -regularity

In this chapter we want to introduce and study the notion of some kind of „regularity“ which sits between regular and F -split Rings. For this we first note the following

Exercise. If for some non-zero divisor $c \in R$ the map $R \rightarrow F_*^e R \in \text{Alg}_{\mathbb{F}_p}$, defined by $1 \mapsto F_*^e c$, is split, then so does $R \rightarrow F_*^{e+1} R, 1 \mapsto F_*^{e+1} c$.

With this in mind the following definition makes sense.

Definition 3.1. Let $R \in \text{Alg}_{\mathbb{F}_p}$ be a noetherian, F -finite ring. Then R is called (*strongly*) F -regular if for all $c \in R \setminus R^0$ and for one, and hence for all e big enough the map

$$\begin{aligned} R &\rightarrow F_*^e R, \\ 1 &\mapsto F_*^e c, \end{aligned}$$

splits. Here $R^0 := \bigcup_{\mathfrak{p} \text{ minimal prime}} \mathfrak{p}$ is the set of all zero-divisors.

This definition can also be cut up by the following

Definition 3.2. Let $R \in \text{Alg}_{\mathbb{F}_p}$ be a noetherian, F -finite ring and $c \in R \setminus R^0$. Then R is called *finely F -split along c* if for one, and hence for all e big enough the map

$$\begin{aligned} R &\rightarrow F_*^e R, \\ 1 &\mapsto F_*^e c, \end{aligned}$$

splits.

With this definition, a ring R is F -regular if and only if R is finely F split along c for all possible c , i.e. for all c which are non-zero-divisors.

Remark 3.3. For $R \in \text{Alg}_{\mathbb{F}_p}$, if there exists some $c \in R \setminus R^0$ such that R is finely F -split along c then R is F -split. This follows directly once one notices that $R \rightarrow F_* R$ sending $1 \mapsto F_* c$ can be factored by the Frobenius as $R \xrightarrow{F^e} F_*^e R \xrightarrow{\cdot F_*^e c} R$.

Theorem 3.4. Let $R \in \text{Alg}_{\mathbb{F}_p}$ be F -finite and noetherian. Then

- i) We have R regular implies R is F -regular implies R is F -split.
- ii) If $R \hookrightarrow S$ is split, then, if S is F -regular, then R is F regular.
- iii) R is F regular if and only if for all $\mathfrak{p} \in \text{spec}(R)$, $R_{\mathfrak{p}}$ is F -regular.
- iv) If $(R, \mathfrak{m})$ is local, then R is F -regular if and only $\widehat{R}$ is F -regular.
- v) The F -regular locus is open in $\text{spec}(R)$
- vi) If R is F split, then R is normal, i.e. integrally closed in it's total field of fractions, and Gorenstein.
- vii) If R is F -regular then for all smooth/regular varieties with a proper birational map $\pi : X \rightarrow R$ we have $H^i(X, \mathcal{O}_X) = 0$ for all $i > 0$ which is equivalent to $R\pi_*\omega_X = \omega_R$.

The last part of the theorem says that R only has rational singularities. However, this part is also over the scope of this course and hence will not be discussed. Instead we will focus on and prove the other parts.

Proof. ii), iii) and iv) are quite obvious, and v) and vi) need more preparations, so here we will only proof the R regular implies that R is F -regular. By the usual procedure we can assume $(R, \mathfrak{m})$ to be local, regular and complete. Let $c \in R \setminus R^0$. We have to show that

$$\mathrm{ev}_{F_*} : \mathrm{Hom}_R(F_*^e R, R) \rightarrow R$$

is surjective for some $e \gg 0$. By the Krull intersection theorem, [Mat89, Theorem 8.9], and since R is complete, we have

$$\bigcap_e \mathfrak{m}^{[p^e]} \subseteq \bigcap_n \mathfrak{m}^n = 0$$

so there exists some $e \gg 0$ such that $c \notin \mathfrak{m}^{[p^e]} = F_*^e \mathfrak{m}^{[p^e]} = \mathfrak{m} F_*^e R$. It follows that $F_*^e c \in F_*^e R \setminus (F_*^e R) \mathfrak{m}$. This is equivalent to $F_*^e c \neq 0 \in F_*^e R / \mathfrak{m} F_*^e R$, which is a $R/\mathfrak{m}$ vector-space. So there exists a basis $(F_*^e c, e_2, \dots, e_n)$ of $F_*^e R / \mathfrak{m} F_*^e R$ for some n , so by the Nakayama-Lemma $F_*^e c$ is an element of a generating system of $F_*^e R$. But using Kunz, F -finiteness and that R is local, $F_* R$ is a free R -module, so we have a projection $\pi : F_*^e R \rightarrow R$ defined as the composition

$$F_*^e R \rightarrow F_*^e c R \cong R$$

which defines a splitting of $R \rightarrow F_*^e R, 1 \mapsto F_*^e c$. □

Here is somewhat of an intuition for what being F -regular means: for R an integral domain, being finely F split along some $c \neq 0$ mean implies that $R \hookrightarrow R^{1/p^e} \xrightarrow{c^{1/p^e}} R^{1/p^e}$ splits for some $e \gg 0$. But c^{1/p^e} is „tiny“ so R is still F split after multiplication by some ϵ . This already goes in the direction of „almost mathematics“ as defined by Professor Ofer Gabber. Next we want to show that the F -regular locus is open in $\mathrm{spec}(R)$. For this we need the following

Theorem 3.5 (Testelement). *Let R be noetherian and F -finite. Let $D(d) := \mathrm{spec}(R) \setminus V(d) \subseteq \mathrm{spec}(R)$ be a standard open set for some $d \in R$ such that $R[d^{-1}]$ is F -regular and R is finely F -split along d . Then R is F -regular.*

Proof. By assumption, R is finely F -split along d and hence, R is F -split. This implies that R is reduced and hence the regular locus of R is non-empty and open. Let $c \in R \setminus R^0$. We want to show that $R \rightarrow F_*^e R \xrightarrow{F_*^e c} F_*^e R$ is split, which is equivalent to $\mathrm{ev}_{F_*^e c} : \mathrm{Hom}_R(F_*^e R, R) \rightarrow R$ is surjective for some $e \gg 0$. By assumption $R[d^{-1}]$ is F -regular, so we know that $\mathrm{ev}_{F_*^e c} : \mathrm{Hom}_{R[d^{-1}]}(F_*^e R[d^{-1}], R[d^{-1}]) \rightarrow R[d^{-1}]$ is surjective for some $e \gg 0$. It follows that for some $t \gg 0$, d^{p^t} is in the image of $\mathrm{ev}_{F_*^e c}$. Let $\psi \in \mathrm{Hom}_R(F_*^e R, R)$ be a preimage. Then, for $\kappa : F_*^t R \rightarrow R$ a Frobenius split, we have

$$F_*^{e+l+t} R \xrightarrow{F_*^{l+t} \psi} F_*^{l+t} R \xrightarrow{F_*^l \kappa} F_*^l R \xrightarrow{\phi} R$$

$$F_*^{e+l+t} c \longmapsto F_*^{l+t} d^{p^t} \longmapsto F_*^l d \longmapsto 1$$

where the last map exists as R is F finely F -split along d . □

This theorem directly implies the following

Corollary 3.6. *Let R be F -finite, noetherian and $d \in R \setminus R^0$ such that $R[d^{-1}]$ is regular and R is finely F -split along d . Then R is F -regular.*

We are no prepared to prove that the F -regular locus is open.

Proof. Let $\mathfrak{m} \in \text{spec}(R)$ be such that $R_{\mathfrak{m}}$ is F -regular. It suffices to show that there exists some standard open $U := \text{spec}(R_g)$ such that $\mathfrak{m} \in U$ and R_g is F -regular. Since $R_{\mathfrak{m}}$ is reduced and excellent, as R is F -finite, there exists an open $U \subseteq \text{spec}(R)$ with $\mathfrak{m} \in U$ such that U is reduced. So by replacing R with some affine subset we can assume that R is reduced. But R being reduced and F -finite implies that the regular locus of R is open and non-empty. Thus there exists some $d \in R \setminus R^0$ such that R_d is regular. If $\mathfrak{m} \notin \mathfrak{m}$ we are done, so assume $d \in \mathfrak{m}$. But then we have that the obstruction of R to be F -regular is, by the testelement theorem (?), exactly the ideal

$$J_e = \text{Im} \left(\text{Hom}_R(F_*^e R, R) \xrightarrow{\text{ev}_{F_*^e d}} R \right) \subseteq R$$

for some $e \gg 0$. In other words for $\mathfrak{n} \in \text{spec}(R)$ we have $R_{\mathfrak{n}}$ is F -regular if and only if $(J_e)_{\mathfrak{n}} = R_{\mathfrak{n}}$ for some $e \gg 0$. But by assumption $R_{\mathfrak{m}}$ is F -regular, so there exists such an e with $(J_e)_{\mathfrak{m}} = R_{\mathfrak{m}}$ which is equivalent to $J_e \not\subseteq \mathfrak{m}$ and hence $\mathfrak{m} \notin V(J_e)$. But conversely all $\mathfrak{n} \notin V(J_e)$ are F -regular points, i.e. $R_{\mathfrak{n}}$ is F -regular, in R , showing the claim. $\square$

Next we show that F -regularity implies normality:

Proof. Assume R to be F -regular. In particular, R is reduced. Write $Q(R)$ for the total quotientring of R , i.e. $Q(R) = S^{-1}R$ with $S = R \setminus R^0$ and $R \subseteq Q(R)$. Let $\frac{x}{y} \in Q(R)$ be integral over R , i.e. there exists $r_i \in R$ such that

$$\left(\frac{x}{y}\right)^n + r_1 \left(\frac{x}{y}\right)^{n-1} + \dots + r_n = 0.$$

It follows from commutative algebra that $R \hookrightarrow R[\frac{x}{y}]$ is finite and that there exists a $c \in R \setminus R^0$ such that $cR[\frac{x}{y}] \subseteq R$, say $c = y^n$. It follows that $c \left(\frac{x}{y}\right)^{p^e} \in R$ for all e and hence $cx^{p^e} \in (y^{p^e})$ for all e , i.e. for all e there exists some $r_e \in R$ such that $cx^{p^e} = r_e y^{p^e}$. Since R is F -regular it follows that there exists a Frobenius split $\varphi : F_*^e R \rightarrow R$, $F_*^e c \mapsto 1$ for some $e \gg 0$. But then

$$x = \varphi(cx^{p^e}) = \varphi(r_e y^{p^e}) = \varphi(r_e)y$$

and hence $\varphi(r_e) = \frac{x}{y} \in R$. $\square$

The last part of theorem 3.4 need the definition of *Cohen-Macaulay* rings.

Definition 3.7. For a local Ring $(R, \mathfrak{m})$ a sequence $x_1, \dots, x_n \in \mathfrak{m}$ is called *regular sequence* if and only if for all i we have that x_i is not a zerodivisor in $R/(x_1, \dots, x_{i-1})$, or equivalently that for all i

$$0 \longrightarrow R/(x_1, \dots, x_{i-1}) \xrightarrow{x_i} R/(x_1, \dots, x_{i-1}) \longrightarrow R/(x_1, \dots, x_i) \longrightarrow 0$$

is exact.

More general for a R -module M we call $x_1, \dots, x_n \in \mathfrak{m}$ a *regular sequence on M* if and only if for all i we have that x_i is not a zerodivisor on $M/(x_1, \dots, x_{i-1})$. We call the length of a maximal regular sequence on M the *depth of M* , written as $\text{depth}_R M$.

It follows from Krull's principle ideal theorem that $\dim R \geq \text{depth}_R M$ for all R -modules M .

At this point it is not at all clear that the notion of the depth is well-defined. This will be resolved later.

Proposition 3.8. *Let $(R, \mathfrak{m})$ be a noetherian local ring. Then the following statements are equivalent.*

- i) There exists a regular sequence on R of length $\dim R$, i.e. $\dim R = \text{depth}_R R$.
- ii) One, and hence all, parameter systems of R is regular sequence.
- iii) The Koszulkomplex of every parametersystem is a resolution of $R/(x_1, \dots, x_n)$.
- iv) The local cohomology of R shifted by $d = \dim R$ is concentrated in 0, i.e. $R\Gamma_{\mathfrak{m}}(R) \simeq H_{\mathfrak{m}}^d(R)[-d]$, i.e. for all $i \neq d$ we have $H_{\mathfrak{m}}^i(R) = 0$.
- v) The dualizing complex is concentrated in one degree.

Definition 3.9. A local noetherian ring $(R, \mathfrak{m})$ is called *Cohen-Macaulay* if R satisfies one and hence all of the equivalent statements of proposition 3.8.

As a short reminder, a parametersystem of R is a sequence $x_1, \dots, x_n$ such that $\sqrt{(x_1, \dots, x_n)} = \mathfrak{m}$.

Remark 3.10. As a short exercise, the interested reader can prove that $(R, \mathfrak{m})$ is Cohen-Macaulay if and only if $(\widehat{R}, \widehat{\mathfrak{m}})$ is Cohen-Macaulay.

Our next step in proving that F -regular rings are Cohen-Macaulay is to prove that if $A \hookrightarrow R$ is a finite extension with A regular, then R is Cohen-Macaulay if and only if R is a locally free A -module. For this we need the *Auslander-Buchsbaum-Serre* formula:

Theorem 3.11 (Auslander-Buchsbaum-Serre). *Let M be a R -module for $(R, \mathfrak{m})$ a noetherian local ring. If the projective dimension $\text{pd}(M) < \infty$ then*

$$\text{depth}_R R - \text{pd} M = \text{depth}_R M$$

We will prove this formula using the following two descriptions of the projective dimension and the depth of M :

$$\begin{aligned} \text{pd} M &= \sup \left\{ i \mid \text{Tor}_i^R(R/\mathfrak{m}, M) \neq 0 \right\} = \sup \left\{ -i \mid (R/\mathfrak{m}) \otimes_R^{\mathbb{L}} M \in \mathcal{D}^{\geq -1}(R) \cap \mathcal{D}^{\leq 0} \right\}, \\ \text{depth}_R M &= \min \left\{ i \mid \text{Ext}_R^i(R/\mathfrak{m}, M) \neq 0 \right\} = \min \left\{ i \mid \mathbb{R}\text{Hom}_R(R/\mathfrak{m}, M) \in \mathcal{D}^{\geq i}(R) \right\}. \end{aligned}$$

We let the reader prove the expression for $\text{pd} M$ for $(R, \mathfrak{m})$ local and noetherian and only prove the expression for the depth.

Remark 3.12. Let $\mathfrak{p} \in \text{Ass} M$ be an associated prime, that is $\mathfrak{p} = \text{Ann}_R m$ for some $m \in M$, or equivalently the R -module morphism $R \rightarrow M, 1 \mapsto m$ factors as $R \twoheadrightarrow R/\mathfrak{p} \hookrightarrow M$, implying that $\text{Hom}_R(R/\mathfrak{p}, M)$ is non-trivial. From commutative algebra it is known that the set of all associated prime ideals of M is finite if R is noetherian, see [Mat89, Theorem 6.5].

Further the union of all associated primes of a module is exactly the set of all zerodivisors on M . Notice further that if $I \subseteq R$ is an Ideal consisting of zerodivisors on M , then $I \subseteq \bigcup_{\mathfrak{p} \in \text{Ass}(M)} \mathfrak{p}$ and hence $I \subseteq \mathfrak{p}$ for some $\mathfrak{p} \in \text{Ass}(M)$.

If now $(R, \mathfrak{m})$ is local noetherian it follows that $\text{depth}_R M = 0$ if and only if $\mathfrak{m} \in \text{Ass}(M)$ if and only if $\text{Hom}_R(R/\mathfrak{m}, M) \neq 0$.

The remark shows that $\text{depth}_R M = \min \left\{ i \mid \text{Ext}_R^i(R/\mathfrak{m}, M) \neq 0 \right\}$ in case that $\text{depth}_R M = 0$.

For the general case we need two lemmata.

Lemma 3.13. i) *If there exists $x \in \text{Ann}_R N$, x a M -regular element, then $\text{Hom}_R(M, N) = 0$.*

ii) *If R is noetherian and M, N are finitely generated R -modules, then the converse is also true*

Proof. For i) let $\varphi \in \text{Hom}_R(M, N)$ and $n \in N$. Then

$$0 = \varphi(0) = \varphi(xn) = x\varphi(n).$$

But x acts injectively on M and hence $\varphi(n) = 0$. As $n \in N$ was arbitrary, $\varphi = 0$.

For ii) assume that there does not exist an M -regular $x \in \text{Ann}_R N = I$. Then we have $I \subseteq \bigcup_{\mathfrak{p} \in \text{Ass}(M)} \mathfrak{p}$ and so by prime avoidance $I \subseteq \mathfrak{p}$ for some $\mathfrak{p} \in \text{Ass}(M) \subseteq \text{Supp} M$. Hence $M_{\mathfrak{p}} \neq 0$. But by construction $N_{\mathfrak{p}} \neq 0$ and we obtain a commutative diagram

$$\begin{array}{ccccc} N_{\mathfrak{p}} & \longrightarrow & \kappa(\mathfrak{p}) & \hookrightarrow & M_{\mathfrak{p}} \\ & \searrow & \uparrow & & \\ & & N_{\mathfrak{p}} \otimes \kappa(\mathfrak{p}) & & \end{array}$$

Hence $N_{\mathfrak{p}} \otimes \kappa(\mathfrak{p}) \neq 0$ and

$$0 \neq \text{Hom}_{R_{\mathfrak{p}}}(N_{\mathfrak{p}}, M_{\mathfrak{p}}) \cong \text{Hom}_R(N, M)_{\mathfrak{p}} \quad (3.0.1)$$

where we used that $S^{-1}(-)$ is flat. But this is only possible if $\text{Hom}_R(M, N) \neq 0$ to begin with showing the claim. $\square$

Lemma 3.14. *Let N, M be R -modules, $x = x_1, \dots, x_n$ a M -regular sequence with $x \subseteq \text{Ann}_R N$. In this case $\text{Hom}_R(N, M/xM) \cong \text{Ext}_R^n(N, M)$.*

Proof. We prove the claim by induction, the case $n = 0$ being obvious. So assume that $\text{Hom}_R(N, M/(x_1, \dots, x_{n-1})M) \cong \text{Ext}_R^{n-1}(N, M)$ for all regular sequences of length $n - 1$. By assumption we know that x_n is regular on $M/(x_1, \dots, x_{n-1})M$. By lemma 3.13 $\text{Hom}_R(N, M/(x_1, \dots, x_{n-1})M) \cong \text{Ext}_R^{n-1}(N, M) = 0$. Further we have an exact sequence

$$0 \longrightarrow M \xrightarrow{x_1} M \longrightarrow M/x_1M \longrightarrow 0.$$

Applying Ext_R^* we obtain a long exact sequence

$$\begin{array}{ccccccc} \dots & \longrightarrow & \text{Ext}_R^{n-1}(N, M) = 0 & \longrightarrow & \text{Ext}_R^{n-1}(N, M/x_1M) & \longrightarrow & \text{Ext}_R^n(N, M) \\ & & & & \searrow^{x_1} & & \\ \text{Ext}_R^n(N, M) & \longleftarrow & \dots & & & & \end{array}$$

But $x_1 \in \text{Ann}_R N$ so the diagonal map is equal to zero and hence $\text{Ext}_R^{n-1}(N, M/x_1M) \cong \text{Ext}_R^n(N, M)^1$. But then

$$\text{Ext}_R^n(N, M) \cong \text{Ext}_R^{n-1}(N, M/x_1M) \cong \text{Hom}_R(N, M/xM)$$

by the induction hypothesis. $\square$

Now the general case for $\text{depth}_R M = \min \{i \mid \text{Ext}_R^i(R/\mathfrak{m}, M) \neq 0\}$ follows easily. We have $\text{Ext}_R^n(R\mathfrak{m}, M) \cong \text{Hom}_R(R/\mathfrak{m}, M/xM) \neq 0$ by part ii) of lemma 3.13 and since x is maximal by assumption, and further $\text{Ext}_R^i(R/\mathfrak{m}, M) = 0$ for all $i < n$ by the proof of lemma 3.14.

We are now ready to proof the Auslander-Buchsbaum-Serre Formula.

¹This is also known as *dimension shifting*.

Proof. Let $(R, \mathfrak{m})$ be noetherian local and M a finite R -module with finite projective dimension. For brevity write $R/\mathfrak{m} = k$. we have

$$\begin{aligned}\mathbb{R}\mathrm{Hom}_R(k, R) \otimes_k^{\mathbb{L}} k \otimes_R^{\mathbb{L}} M &\cong \mathbb{R}\mathrm{Hom}_R(k, R) \otimes_R^{\mathbb{L}} M \\ &\cong \mathbb{R}\mathrm{Hom}_R(k, R \otimes_R^{\mathbb{L}} M) \\ &\cong \mathbb{R}\mathrm{Hom}_R(k, M)\end{aligned}$$

Here we used in the second isomorphism that $\mathrm{pd} M < \infty$, which implies that we can find a finite free resolution of M by finite free R -modules as $(R, \mathfrak{m})$ is local noetherian and M is finitely generated.

Now notice² that $\mathbb{R}\mathrm{Hom}_R(k, R) \in \mathcal{D}^{\geq \mathrm{depth}_R R}$, $k \otimes_R^{\mathbb{L}} M \in \mathcal{D}^{\geq -\mathrm{pd} M}$, hence $\mathbb{R}\mathrm{Hom}_R(k, R) \otimes_k^{\mathbb{L}} k \otimes_R^{\mathbb{L}} M \in \mathcal{D}^{\mathrm{depth}_R R - \mathrm{pd} M}$ as well as $\mathbb{R}\mathrm{Hom}_R(k, M) \in \mathcal{D}^{\geq \mathrm{depth} M}$ by the lemmas we proved above, showing the claim. $\square$

Exercise. For a noetherian local ring $(R, \mathfrak{m})$ we have the following inequality

$$\mathrm{depth}_R R = \min \{ \dim R/\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Ass} R \} \leq \max \{ \dim R/\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Ass} R \} = \dim R.$$

In particular if R is Cohen-Macaulay then R is equidimensional and has no embedded dimension.

We can now prove

Proposition 3.15. *Let $A \hookrightarrow R \in \mathrm{Ring}$ be finite, A regular. Then R is Cohen-Macaulay if and only if R is locally a free A -module.*

Proof. As the statement is local we can assume WLOG that A is local. Then we have by Auslander-Buchsbaum-Serre

$$\infty > \mathrm{pd}_A R = \mathrm{depth}_A A - \mathrm{depth}_A R.$$

Since A is regular, $\mathrm{depth}_A A = \dim A$ and since $A \hookrightarrow R$ is finite, $\mathrm{depth}_A R = \mathrm{depth}_R R$ as well as, using going up, $\dim A = \dim R$. As $\mathrm{depth}_R R \leq \dim R$ we get from rewriting Auslander-Buchsbaum-Serre

$$\mathrm{pd}_A R = \dim R - \mathrm{depth}_R R \geq 0$$

and equality holds if and only if $\mathrm{depth}_A A = \mathrm{depth}_A R$. In other words, R is Cohen-Macaulay, by definition if and only if $\mathrm{depth}_R R = \dim R$, if and only if $\mathrm{pd}_A R = 0$ if and only if R is a free A -module. $\square$

Our next goal is to proof that being F -regular implies the CM property. For this we need the following Lemma

Lemma 3.16. *Let $B \hookrightarrow R$ be a finite integral ring extension of integral domains. then there exists $\varphi \in \mathrm{Hom}_B(R, B)$ such that $\varphi(1) \neq 0$.*

Proof. The assumptions assure that $L := \mathrm{Frac}(B) \hookrightarrow \mathrm{Frac}(R) = L \otimes_B R$ is a finite field extension. This implies that

$$L \otimes_B \mathrm{Hom}_B(R, B) = \mathrm{Hom}_L(L \otimes_B R, L).$$

²It might be helpful drawing the where the derived objects live.

Hence we get the following commutative diagram

$$\begin{array}{ccccc}
R & \hookrightarrow & R \otimes_B L & \xrightarrow{\psi} & L \\
\uparrow & & \psi|_R & & \uparrow \\
B & \xrightarrow{\text{id}} & B & & B
\end{array}$$

The image of R under $\psi|_R$ is integral over $B \hookrightarrow L$. Hence there exists some $c \in B \setminus \{0\}$ such that $c\psi|_R(R) \subseteq B$ finishing the proof. $\square$

Proposition 3.17. *If R is F -regular then R is CM.*

Proof. WLOG we can assume $(R, \mathfrak{m})$ to be a complete, local, F -regular noetherian ring. This implies that R is an integral domain.

Let $x_1, \dots, x_d$ be a parameters system with $d = \dim(R)$. We have to show that this system is a regular sequence. So assume the contrary. Then there exists some $z \in R$ such that $zx_i \in (x_1, \dots, x_{i-1})$ and $z \in (x_1, \dots, x_{i-1}) \setminus \{0\}$. Thus we have for all e an equation of the form

$$z^{p^e} x^{p^e} = r_1 x_1^{p^e} + \dots + x_{i-1}^{p^e}$$

for $r_i \in R$ depending on e . Let $A = K[[x_1, \dots, x_n]]$ where $K = R/\mathfrak{m}$. As A is a powerseries ring, A is regular and hence CM. Further $A \hookrightarrow R$ is finite. Let F be the integral equation for z . Then we have $A \subseteq A[T]/f(T) =: B \subseteq R$.

In fact, B is CM. For this we need to show that $\dim(B) = \text{depth}(B)$. For this consider $S = A[T]_{(\mathfrak{m}, T)}$. Then there is a short exact sequence of S modules.

$$0 \longrightarrow S \xrightarrow{f} S \longrightarrow S/f \longrightarrow 0.$$

Applying $\text{Ext}_S^*(S/\mathfrak{m}f, -)$ we obtain a long exact sequence of Ext-modules. In this sequence, the multiplication by f induces the 0 map, as f is a zerodivisor in $S/\mathfrak{m}$. SO the long exact sequence splits into short exact sequences and we see that

$$\text{depth}(S/f) = \text{depth}(S) - 1 = \dim(S) - 1 = \text{depth}(S/f)$$

showing that $B = S/f$ is CM.

Now let $\varphi \in \text{Hom}_B(R, B)$ be as in the Lemma. Applying φ to the equation for z we obtain

$$cz^{p^e} x_i^{p^e} = \varphi(z^{p^e} x_i^{p^e}) = \varphi(r_1) x_1^{p^e} + \dots + \varphi(r_{i-1}) x_{i-1}^{p^e}.$$

But we also know that $x_1^{p^e}, \dots, x_d^{p^e}$ is a parameter system in B , as it is one in A and $A \subseteq B$ is integer. But B is CM, so $x_1^{p^e}, \dots, x_d^{p^e}$ is in fact a regular sequence, so the above equation implies

$$cz^{p^e} \in (x_1^{p^e}, \dots, x_{i-1}^{p^e}).$$

Thus we can write

$$cz^{p^e} = a_1 x_1^{p^e} + \dots + a_{i-1} x_{i-1}^{p^e}.$$

This equation now gives us the desired contradiction:

As R is F -regular there exists $\psi \in \text{Hom}_R(F_*^e R, R)$ such that $\psi(F_*^e c) = 1$ for e great enough. Applying ψ to the above equation then gives

$$z = \psi(cz^{p^e}) = \psi(a_1) x_1 + \dots + \psi(a_{i-1}) x_{i-1} \in (x_1, \dots, x_{i-1}).$$

This finishes the proof. $\square$

The dependencies of notions are now the following

$$\begin{array}{ccccc}
 \text{Regular} & \implies & F\text{-regular} & \implies & F\text{-split} \\
 & & \Downarrow & & \Downarrow \\
 & & \text{CM and normal} & & \text{reduced}
 \end{array}$$

In particular, we get the following

Corollary 3.18 (Hochster-Roberts 1974). *A direct summand of a regular ring in characteristic $p > 0$ is CM.*

Proof. If A is a direct summand of a regular ring R then A is F -regular and hence CM. $\square$

This ends our study of F -regularity. In the next and final chapter we will introduce another type of singularities, called *F -rational* singularities. They are much more „local in nature“ and we will need the aperture of *local cohomology*.

4 F -rational

As mentioned in the previous chapter we will start by introducing local cohomology. In particular we will discuss the induced action of the Frobenius on local cohomology.

Definition 4.1. Let R be a ring and $I \subseteq R$ be an ideal. Consider the I -torsion functor $\Gamma_I : \text{Mod}_R \rightarrow \text{Mod}_R$ defined by

$$\begin{aligned}\Gamma_I M &:= \left\{ m \in M \mid \exists i : I^i m = 0 \right\} \\ &= \bigcup_i \text{Ann}_M I^i \\ &= \bigcup_e \text{Ann}_M I^{[p^e]}\end{aligned}$$

If $I = (x_1, \dots, x_n)$ is finitely generated, we further obtain an equality

$$\Gamma_I M = \ker \left(M \rightarrow \bigoplus M_{x_i} \right).$$

Then *local cohomology* is defined to be the right derived functor of Γ_I , written $\mathbb{R}\Gamma_I$.

In more geometric terms, local cohomology can be imagined as cohomology on the affine spectrum $\text{spec}(R)$ with support in the closed subscheme $\text{spec}(R/I)$.

Example 4.2. Take $(R, \mathfrak{m})$ some local ring. Let $M = R$, $I = \mathfrak{m} = (x_1, \dots, x_n)$. Then local cohomology $\mathbb{R}\Gamma_{\mathfrak{m}} R$ is given by a Čech complex

$$\mathbb{R}\Gamma_{\mathfrak{m}} R \cong \left(R \rightarrow \bigoplus R_{x_i} \rightarrow \bigoplus_{i < j} R_{x_i x_j} \rightarrow \dots \rightarrow R_{x_1 \dots x_n} \right)$$

We will denote the cohomology groups of the local cohomology complex as

$$H_I^i(-) := \mathbb{R}\Gamma_I^i(-) := H^i(\mathbb{R}\Gamma_I(-)).$$

As $\mathbb{R}\Gamma_I(-)$ is a functor, when applied to a local $\mathbb{F}_p$ -algebra $(R, \mathfrak{m})$ we obtain a natural action of the Frobenius on local cohomology

$$F^e : \mathbb{R}\Gamma_{\mathfrak{m}} R \rightarrow \mathbb{R}\Gamma_{\mathfrak{m}}(F_*^e R) \cong F_*^e \mathbb{R}\Gamma_{\mathfrak{m}} R.$$

As stated in example 1.8, F_* is an exact functor, hence we get an isomorphism $H_I^i F_*^e \cong F_*^e H_I^i$. We will (suggestively) write $F_*^e \eta^{p^e}$ for the image of the induced map $F^e : H_{\mathfrak{m}}^i(R) \rightarrow F_*^e H_{\mathfrak{m}}^i(R)$

Exercise. 1) Let $(R, \mathfrak{m})$ be local. Then R is CM if and only if $H_{\mathfrak{m}}^i(R) = 0$ for all $i < d = \dim R$, or equivalently $\mathbb{R}\Gamma_{\mathfrak{m}} R$ is a R -module in degree d .

(Hint: Show that CM is equivalent to $\text{Ext}_R^i(R/\mathfrak{m}, R) = 0$ for all $i < d$ and then show that the limit of all ext groups is exactly local cohomology, using that these Ext-groups characterize torsion.) See [Sta24, 0AVZ] for the solution.

2) For $(R, \mathfrak{m})$ complete and normal there exists $c \in R \setminus R^0$ such that $cH_{\mathfrak{m}}^i(R) = 0$ for all $i \neq \dim R$.

The next example gives another proof that F -regularity implies CM, assuming the exercises above.

Example 4.3. Let R be F -regular, c as in exercise 2). Then we have the following diagram

$$\begin{array}{ccccccc} R & \xrightarrow{F^e} & F_*^e R & \xrightarrow{F_*^e c} & F_*^e R & \xrightarrow{\pi} & R \\ & & & & & \searrow & \\ & & & & & & id_R \end{array}$$

We can now apply local cohomology groups for $i \neq \dim(R)$ to obtain a factorization of the identity by multiplication with $F_*^e c$ for any $c \in R \setminus R^0$. By the previous exercise this map is zero for some such c , hence $H_{\mathfrak{m}}^i(R) = 0$ for all $i \neq \dim(R)$.

Local cohomology, as we defined it, is inherently quasi isomorphism invariant. Hence, for $(R, \mathfrak{m})$ a local noetherian ring with $\mathfrak{m} = (f_1, \dots, f_d)$ we can consider local cohomology as a functor on the derived category of R -modules. In this case it turns out that local cohomology is in fact a coreflexive colocalization:

$$\mathbb{R}\Gamma_{\mathfrak{m}}(-): \mathcal{D}(R) \xrightarrow{\quad \perp \quad} \mathcal{D}_{\mathfrak{m}^\infty}(R)$$

Here $\mathcal{D}_{\mathfrak{m}^\infty}(R)$ is the full subcategory containing $\mathfrak{m}^\infty$ torsion. In particular, if any $f \in \mathfrak{m}$ acts invertible on M then $\mathbb{R}\Gamma_{\mathfrak{m}} M = 0$

The explicit description using the Čech complex can be transferred to the derived setting. Let $M \in \mathcal{D}(R)$.

$$\mathbb{R}\Gamma_{\mathfrak{m}} M = M \otimes_R^{\mathbb{L}} \left(R \rightarrow \prod_i R_{f_i} \rightarrow \prod_{i < j} R_{f_i f_j} \rightarrow \dots \rightarrow R_{f_1 \dots f_d} \right)$$

We also have another “explicit” description using homotopy colimits.

$$\begin{aligned} \mathbb{R}\Gamma_{\mathfrak{m}} M &= \operatorname{hocolim}_i \left(\mathbb{R}\operatorname{Hom}_R \left(R/\mathfrak{m}^i, M \right) \right) \\ H_{\mathfrak{m}}^n(M) &= \operatorname{colim}_i \left(\operatorname{Ext}_R^n \left(R/\mathfrak{m}^i, M \right) \right). \end{aligned}$$

In particular $\operatorname{depth} M = \min \{n \mid H_{\mathfrak{m}}^n(M) = 0\}$.

We can use local cohomology to define new types of singularities.

Definition 4.4. Let $(R, \mathfrak{m})$ be a local $\mathbb{F}_p$ -algebra. Then

- 1) R is called *F-injective* by definition if and only if

$$F^e : H_{\mathfrak{m}}^i R \rightarrow F_*^e H_{\mathfrak{m}}^i(R)$$

is injective for all e and i .

- 2) R is called *F-rational* by definition if and only if R is CM and for all $c \in R \setminus R^0$

$$F^e : H_{\mathfrak{m}}^i R \xrightarrow{F_*^e c \circ F^e} F_*^e H_{\mathfrak{m}}^i(R)$$

is injective for all e .

Exercise. Let $(R, \mathfrak{m})$ be local.

- 1) If R is F -split, then R is F -injective.
- 2) If R is F -regular, then R is F -rational.

- 3) If R is F -rational, then R is F -injective.
- 4) R is F -injective if and only if $\hat{R}$ is F -injective.
- 5) If $\hat{R}$ is F -rational, then R is F rational. (The converse is true if R is F -finite.)

In particular, we have the following image of dependencies

$$\begin{array}{ccccccc}
 \text{Regular} & \implies & F\text{-regular} & \implies & F\text{-rational} & \implies & \text{CM and normal} \\
 & & \Downarrow & & \Downarrow & & \\
 & & F\text{-split} & & F\text{-injective} & &
 \end{array}$$

Further, F -rational implies F -regular, and F -injective implies F -split, the ring under consideration R is (local) Gorenstein, i.e. R is CM and $H_{\mathfrak{m}}^i(R) \cong E_{R/\mathfrak{m}}$. Here E is the injective hull of the residue field $R/\mathfrak{m}$.

Before we continue our adventure in the world of local cohomology, we make a short detour into the theory of the tight closure as introduced by Hochster and Hureke in the eighties and nineties of the last century. The notions of F -injective and F -rational was originally (**True?**) defined in the context of this theory, and it was only later that it was realized, these two notions can also be stated in the formulation of local cohomology.

In the theory of the tight closure (and also for local duality), the notion of an injective hull is quite important. For this reason, we will first review the definition.

Definition 4.5. Let R be a ring, M a R -module. Then $N \hookrightarrow M \in \text{Mod}_R$ is called *essential* by definition if and only if for all submodule $M' \subseteq M$ we have that $N \cap M' \neq 0$ if and only if for all $x \in M$ there exists some $r \in R$ such that $rx \neq 0$ and $rx \in N$.

Another important definition which we also already used in the preceding paragraph is the notion of an injective hull.

Definition 4.6. 1) Let R be a ring. An R -module E is called *injective* by definition if and only if every lifting problem

$$\begin{array}{ccc}
 N & \hookrightarrow & M \\
 \downarrow & \nearrow & \downarrow \\
 E & \longrightarrow & 0
 \end{array}$$

there exists a (in general non-unique) solution, indicated by the dotted arrow¹. This is equivalent to $\text{Hom}_R(-, E)$ being an exact endofunctor of R -modules.

- 2) Let R be a ring, $M \in \text{Mod}_R$. Then $M \hookrightarrow E(M) \in \text{Mod}_R$ is called an *injective hull* of M if $M \hookrightarrow E(M)$ is essential and $E(M)$ is injective.

We write E instead of $E(M)$ if the module M is clear from the context.

Example 4.7. Here are some examples of injective hulls.

- Let k be a field. Then $E(k) = k$.
- Let $R = k[X]$. Then $E(R/\mathfrak{m}) = H_{\mathfrak{m}}^1(R) = \text{coker}(R \rightarrow R_{\mathfrak{m}})$.

The following exercise gives some intuition for injective modules. See [Sta24, paragraph 3, 47.3] for solutions.

Exercise. Let R be a ring.

¹By Baer's theorem it suffices to check the lifting problem for all ideals $I \subseteq R$.

- 1) If $\{E_i\}_{i \in I}$ is a collection of injective R -modules, then $\prod_{i \in I} E_i$ is injective.
- 2) Assume $R \rightarrow S \in \text{Ring}$ is flat. If E is an injective R -module then $S \otimes_R E$ is an injective S -module. In particular $S^{-1}E$ is an injective $S^{-1}R$ module for any multiplicative set $S \subset R$.
- 3) If $R \rightarrow S \in \text{Ring}$ is an epimorphism, E an S -module such that E is injective as an R -module via the epimorphism, then E is an injective S -module.
- 4) Let $R \xrightarrow{\varphi} S \in \text{Ring}$, E be an injective E -module. Then $\varphi^b E = \text{Hom}_R(S, E)$ is an injective S -module.

This lemma gives even more intuition for what an injective module is, however we will not prove this.

Lemma 4.8. *Let R be a ring. Let $E \subseteq I \in \text{Mod}_R$ such that I is injective. Then E is injective if and only if for every essential intermediate extension $E \subseteq E' \subseteq I$ then $E = E'$.*

This means that E is injective if and only if E has no non-trivial essential extensions. For noetherian rings we have the following additional properties:

- 1) If $\{E_i\}_{i \in I}$ is a collection of injective R -modules, then $\bigoplus_{i \in I} E_i$ is injective.
- 2) If E is injective then for all $f \in R$, $E[f^\infty] = \Gamma_{(f)} E$ is injective. Analogously $\Gamma_J E$ is injective for any ideal $J \subseteq R$.

An important question is whether every Modul has an injective hull. We can answer this question affirmative:

Lemma 4.9. *let R be a ring. then every R -module M has an injective hull. The hull is further unique up to non-unique isomorphism.*

Sketch. Let $M \subset I$ be injective. Consider the set of all essential sub-extensions $M \subseteq M' \subseteq I$. Using Zorns lemma there exists a maximal sub extension $M \subseteq E \subseteq I$. Using the lemma it follows that M' is injective and hence E is an injective hull.

Regarding the uniqueness, consider the following diagram

$$\begin{array}{ccc} N & \xrightarrow{\varphi} & M \\ \downarrow & & \downarrow \\ E_N & \xrightarrow{\exists \varphi'} & E_m \end{array}$$

If φ is injective, so is φ' and if φ is essential, φ' is an isomorphism. In particular, if φ is an isomorphism, so is φ' . $\square$

Remark 4.10. One can show the following:

If we have a diagram

$$\begin{array}{ccc} M & \hookrightarrow & I \\ \downarrow & \nearrow & \\ E_M & \xrightarrow{\text{split}} & \end{array}$$

$M \hookrightarrow I \in \text{Mod}_R$ with I injective. That is $M \subseteq E_M \oplus I' = I$. That is every injective module over a module contains the injective hull as a direct summand.

Definition 4.11. Let $R \in \text{Alg}_{\mathbb{F}_p}$ and $I \subseteq R$ some ideal. Then the *tight closure* I^* of I is defined as the set of all $y \in R$ such that there exists some fixed $c \in R \setminus R^0$ and $e_0 \geq 0$ with $cy^{p^e} \in I^{[p^e]}$ for all $e \geq e_0$.

The intuition is that after taking p^e -th roots, if this would be possible, then $c^{1/p^e}y \in I^{1/p^e}$, so y is „almost“ in I .

Exercise. Let $R \in \text{Alg}_{\mathbb{F}_p}$ and $I \subseteq R$ some ideal.

- 1) $I \subset I^*$.
- 2) $(I^*)^* = I^*$.
- 3) $I \subseteq J \Rightarrow I^* \subseteq J^*$.
- 4) $\sqrt{0} \subseteq 0^*$.

The definition of the tight closure works similarly for modules.

Definition 4.12. Let $R \in \text{Alg}_{\mathbb{F}_p}$, $M \in \text{Mod}_R$ and $N \subseteq M$ be some submodule. We define the tight closure of N in M , $N_M^* \subseteq M$, as $m \in N_M^*$ by definition if and only if there exists some $c \in R \setminus R^0$ such that $F_*^e c \otimes m \in N^{[p^e]} \subseteq F_*^e R \otimes M$ for all e great enough.

Example 4.13. We have $0_M^* = \{m \in M \mid \exists c : F_*^e c \otimes m = 0 \ \forall e \gg 0\}$.

Proposition 4.14. Let $(R, \mathfrak{m})$ be a local noetherian ring.

- 1) R is F -regular if and only if $0_E^* = 0$ for all injective hulls E of $R/\mathfrak{m}$. This implies that for all ideals $I \subseteq R$ we have $I = I^*$.
- 2) R is F -rational if and only if $0_{H_{\mathfrak{m}}^d(R)}^* = 0$ for $d = \dim(R)$. This is further equivalent to the fact that all ideals $I \subseteq R$ that are generated by a part of a parametersystem, $I^* = I$.

A ring that satisfies the last implication of 1) is called „weak F -regular“.

Proof. We will only proof 1). We start with the fact that F -regularity implies weak F -regularity.

Let $z \in I^*$, that is there exists $c \in R \setminus R^0$ such that for all $e \gg 0$ we have $zF_*^e c = F_*^e(cz^{p^e}) \in IF_*^e R$. This implies $cz^{p^e} = \sum_{i=1}^n r_i x_i^{p^e}$ for $r_i \in R$ and x_i the generators of I . As R is F -regular, there exists a split of $R \rightarrow F_*^e R, 1 \mapsto F_*^e c$ for $e \gg 0$. Let $\pi : F_*^e R \rightarrow R$ be such a splitting. Then

$$z = \pi(zF_*^e c) = \pi\left(\sum_i r_i x_i^{p^e}\right) = \sum_i \pi(r_i) x_i \in I.$$

This shows the claim.

Next we show the equivalence in 1).

Assume first that R is F -regular and $0_E^* \neq 0$, that is, there exists $\eta \in 0_E^* \setminus \{0\}$ such that $F_*^e c \otimes \eta = 0$ for some $c \in R \setminus R^0$ and for all $e \gg 0$ in $F_*^e R \otimes E$. By assumption there exists some $e \gg 0$ such that $R \rightarrow F_*^e R, 1 \mapsto F_*^e c$. Applying the functor $- \otimes_R E$ we obtain that $R \otimes_R E \cong E \rightarrow F_*^e R \otimes_R E, \eta \mapsto F_*^e c \otimes \eta$ is split, a contradiction. This shows “ $\Rightarrow$ ”.

For “ $\Leftarrow$ ” consider $c \in R \setminus R^0$. We will use the following fact which we leave as an exercise for the reader.

Let $(R, \mathfrak{m})$ be a local noetherian ring and M an R -module containing R . Then $R \hookrightarrow M$ is split if and only if for all N the inclusion $N \hookrightarrow M \otimes_R N$ is injective (i.e. $R \hookrightarrow M$ is pure) if and only if $R \otimes_R E \cong E \rightarrow M \otimes_R E$ is injective.

This lemma implies the $R \rightarrow F_*^e R, 1 \mapsto F_*^e c$ is split for $e \gg 0$ if and only if $E \rightarrow F_*^e R \otimes_R E, \eta \mapsto F_*^e c \otimes \eta$ is injective for E the injective hull of the residue field $R/\mathfrak{m} = \kappa$.

We have a commutative diagram

$$\begin{array}{ccc} \kappa & \xrightarrow{\quad} & E \\ & \searrow & \uparrow \\ & & E[\mathfrak{m}] \end{array}$$

where $\kappa \hookrightarrow E$ is essential (for those not familiar with essential extensions, we will define this in the next section, see definition 4.5) and $E[\mathfrak{m}]$ is the socle of E .

As $E[\mathfrak{m}]$ is a submodule of E and we have an injection $\kappa \rightarrow E[\mathfrak{m}]$ this injection is also essential. κ is a field and $E[\mathfrak{m}]$ is a κ vectorspace, but the only essential extension of κ are one dimensional vectorspaces. Thus $\kappa \cong E[\mathfrak{m}]$, that is, the socle of E is one-dimensional, generated by $\eta \in E[\mathfrak{m}]$ (Note, E is not a κ vector space in general, so E can be bigger than $E[\mathfrak{m}]$).

Since $E[\mathfrak{m}] \hookrightarrow E$ is essential it suffices to show that $F_*^e c \otimes \eta \neq 0$. But this is clear as else $F_*^e c \otimes \eta = 0$ for all $e \gg 0$, that is $\eta \in 0_E^*$ which is simply 0 by assumption. $\square$

This ends our short detour into the world of tight closure. We will now return to local cohomology. Our main goal is to prove the local duality theorem, and our first step towards this goal is the Matlis duality. For this we need the following facts for injective hulls.

Lemma 4.15. *Let R be a ring and let $E \in \text{Mod}_R$ be an indecomposable and injective R -module. Then*

- 1) E is an injective hull for ever non-zero submodule.
- 2) Every two submodules of E have a non-zero intersection.
- 3) The set of endomorphisms of E , $\text{End}_R(E)$ is a non-commutative local ring.
- 4) The set of zero divisors on E form a prime ideal of R , $\mathfrak{p}$, and E is a $R_{\mathfrak{p}}$ -module.

Let now E be the injective hull of $R/\mathfrak{m}$. Then

- 1) E is indecomposable.
- 2) E is an injective hull of $\text{Frac}(R/\mathfrak{p}) = \kappa(\mathfrak{p})$.
- 3) E is the injective hull of $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} = \kappa(\mathfrak{p})$ as an $R_{\mathfrak{p}}$ -module.

If R is noetherian then we have the following two additional properties:

- 1) E is indecomposable and an injective R -module if and only if $E = E(R/\mathfrak{p})$ for some $\mathfrak{p} \in \text{spec}(R)$.
- 2) $E = E(R/\mathfrak{m}) = \bigoplus_{\mathfrak{p} \in \text{spec}(R)} E(R/\mathfrak{p})$

Hier Aufzeichnung unklar: Es steht da “Sei R ab jetzt artinsch”, und in der nächsten zeile direkt “Ab hier ist R nicht mehr artinsch”.

To describe the duality theorem we need some notion of a “dual”. This notion is the following.

Definition 4.16. Let $(R, \mathfrak{m})$ be a local ring and M be a finitely generated R -module and let $E = E(R/\mathfrak{m})$ be the injective hull of the residue field. Then the *Matlis-dual* of M is the module $M^\vee := \text{Hom}_R(M, E)$.

Lemma 4.17. *Let $(R, \mathfrak{m})$ be an artinian local ring and M a finitely generated R -module. Then*

$$\text{length}(M) = \text{length}(M^\vee).$$

In particular, $E = \text{Hom}_R(R, E)$ has finite length and is finitely generated.

Proof. We have, as stated previously, $\kappa = R/\mathfrak{m} \cong E[\mathfrak{m}] \subseteq E$ But then $l(R/\mathfrak{m}) = 1$ and $\text{Hom}_R(R/\mathfrak{m}, E) \cong E$. But then the assumption holds for $M = \kappa$. The general assumption now follows by induction using that E is injective, so $\text{Hom}_R(-, E)$ is exact. $\square$

Corollary 4.18. *The natural map*

$$M \rightarrow \text{Hom}_R(\text{Hom}_R(M, E), E), m \mapsto (\psi \mapsto \psi(m))$$

is always injective and if M has finite length, it is an isomorphism.

Proof. This follows directly from the following commutative diagram

$$\begin{array}{ccc} R & \longrightarrow & R/\mathfrak{m} \\ \downarrow & & \downarrow \\ M & \xrightarrow{\exists \varphi} & E(R/\mathfrak{m}) \end{array}$$

□

Exercise. Let $(R, \mathfrak{m})$ be a local artinian ring. Let $I \subseteq R$ be an ideal.

1) $(M/IM)^\vee =: D(M/IM) = D(M)[I] := M^\vee[I].$

2) If $R \xrightarrow{\varphi} S \in \text{Ring}$ is surjective, $E = E(R/\mathfrak{m})$, then

$$E(S/\mathfrak{m}_S) = \varphi^\flat(E(R/\mathfrak{m})) = \text{Hom}_R(S, E(R/\mathfrak{m})) = E(R/\mathfrak{m})[I].$$

3) If M is finitely generated R -module consisting of $\mathfrak{m}^\infty$ -torsion, then we have the following commutative diagram

$$\begin{array}{ccc} M[\mathfrak{m}] & & \\ \downarrow & \searrow & \\ M & \xrightarrow{\exists} & \bigoplus E(R/\mathfrak{m}) \end{array}$$

and hence M is injective.

4) $E(R/\mathfrak{m}) \cong E(\hat{R}/\hat{\mathfrak{m}})$

5) $E(R/\mathfrak{m})$ has DCC.

Hint for 4). Use $R^{\vee\vee} = \text{Hom}_R(E, E) = \lim \text{Hom}_R(E[\mathfrak{m}^n], E[\mathfrak{m}^n]) = \lim (R/\mathfrak{m}^n) = \hat{R}.$

Before we can state the Matlis duality theorem we give a short reminder

Remark 4.19. Let $(R, \mathfrak{m})$ be a noetherian local ring. Then

1) M is finitely generated if and only if M has the ACC if and only if there exists a presentation $R^m \rightarrow R^n \rightarrow M \rightarrow 0.$

2) $\Gamma_{\mathfrak{m}}M = M$ und $M[\mathfrak{m}]$ has $\dim_{\kappa}(M[\mathfrak{m}]) < \infty$ if and only if M has the DCC if and only if there exists a presentation $0 \rightarrow M \rightarrow E^n \rightarrow E^m.$

Corollary 4.20 (Matlis-duality). *Let $(R, \mathfrak{m}, \kappa)$ be a complete noetherian local ring, $E = E(R/\mathfrak{m})$. Then $(-)^\vee = D(-) = \text{Hom}_R(-, E)$ induces an anti equivalence of categories*

$$\{R\text{-moduls with DCC}\} \leftrightarrow \{R\text{-moduls with ACC}\}.$$

Proof. By the exercise we have $R \cong \hat{R} \cong \text{Hom}_R(E, E) = E^\vee$ and of course $E \cong \text{Hom}_R(R, E) = R^\vee$. Thus the statement follows from the previous remark. □

For a general ring R we have the following diagram

$$\begin{array}{ccc} R\text{-DCC} & \xrightarrow{(-)^\vee} & \hat{R}\text{-ACC} \\ & \nwarrow (-)^\vee & \uparrow \widehat{(-)} \\ & & R\text{-ACC} \end{array}$$

Another ingredient needed for local cohomology is the theory of derived completion. We will only define this in the context of local noetherian ring, be aware though, that this can be defined in a much more general setting.

Definition 4.21. Let $(R, \mathfrak{m})$ be a local noetherian ring, $\mathfrak{m} = (f_1, \dots, f_d)$. Then *Derived completion* is the functor

$$\begin{aligned} (-)^{\wedge \mathfrak{m}} : \mathcal{D}(R) &\rightarrow \mathcal{D}_{\mathfrak{m}\text{-comp}}(R) \\ M &\mapsto \mathbb{R}\mathrm{Hom}_R \left(\left(R \rightarrow \prod R_{f_i} \rightarrow \dots \rightarrow R_{f_1 \dots f_d} \right), M \right) \end{aligned}$$

By definition, it seems as the derived completion is tightly related with local cohomology, and this is in fact true.

Exercise. Let $(R, \mathfrak{m})$ be a local noetherian ring, $M \in \mathcal{D}(R)$. Proof the following

$$\begin{aligned} \mathbb{R}\Gamma_{\mathfrak{m}} M^{\wedge \mathfrak{m}} &= \mathbb{R}\Gamma_{\mathfrak{m}} M, \\ (\mathbb{R}\Gamma_{\mathfrak{m}} M)^{\wedge \mathfrak{m}} &= M^{\wedge \mathfrak{m}}. \end{aligned}$$

The idea is to consider the exact triangle $M = \mathbb{R}\mathrm{Hom}_R(R, M) \rightarrow M^{\wedge \mathfrak{m}} \rightarrow C \xrightarrow{+1}$.

As an immediate consequence we get the following

Corollary 4.22. *There is a commutative diagram of categories*

$$\begin{array}{ccc} \mathcal{D}_{\mathfrak{m}^\infty} & \xrightleftharpoons[\mathbb{R}\Gamma_{\mathfrak{m}}]{(-)^{\wedge \mathfrak{m}}} & \mathcal{D}_{\mathfrak{m}\text{-comp}}(R) \\ & \nwarrow \mathbb{R}\Gamma_{\mathfrak{m}} \quad \nearrow (-)^{\wedge \mathfrak{m}} & \\ & \mathcal{D}(R) & \end{array}$$

such that the horizontal arrows form an equivalence of categories.

The last thing we need is the notion of a dualizing complex.

Definition 4.23. Let R be a noetherian ring. A complex $\omega_R \in \mathcal{D}(R)$ is called a *dualizing complex* by definition if and only if one of the following equivalent statements hold.

- 1) ω_R is isomorphic in $\mathcal{D}(R)$ to a bounded complex of injective modules.
- 2) $H^i(\omega_R)$ is finitely generated. (Such a complex is called *coherent*, written $\omega_R \in \mathcal{D}_{\mathrm{coh}}(R)$.)
- 3) $R \rightarrow \mathbb{R}\mathrm{Hom}_R(\mathbb{R}\mathrm{Hom}_R(R, \omega_R), \omega_R) \in \mathcal{D}(R)$ is an isomorphism.

We will write $\mathbb{D}$ for the endofunctor $\mathbb{R}\mathrm{Hom}_R(-, \omega_R)$.

Theorem 4.24. *let R be a noetherian ring. Then*

- 1) $\mathbb{D}$ preserves $\mathcal{D}_{\mathrm{coh}}(R)$.

2) $\mathbb{D}$ induces an anti-auto-equivalence of $\mathcal{D}_{coh}(R)$. This descends to an anti-auto-equivalence of the bounded derived category, and to an anti-equivalence $(\mathcal{D}_{coh}^+(R))^{\text{op}} \simeq \mathcal{D}_{coh}^-(R)$

Proof. Just a short note regarding b). There exists a canonical map

$$K = \mathbb{R}\text{Hom}_R(\omega_R, \omega_R) \otimes_R^{\mathbb{L}} K \rightarrow \mathbb{R}\text{Hom}_R(\mathbb{R}\text{Hom}_R(K, \omega_R), \omega_R)$$

and this is an isomorphism on the categories under consideration. $\square$

Remark 4.25. Of course one could ask whether dualizing complexes exist at all. Without being to specific, they do. And they do so quite often. Here are some examples

- If $R = k$ a field, then $\omega_R = k$.
- If $R = k[X_1, \dots, X_n]$ then $\omega_R = R$ and $\omega_R = \Lambda^n \Omega_{R/k}^1[n]$ are both dualizing complexes.
- if $\varphi : R \rightarrow S \in \text{Ring}$ is finite and ω_R a dualizing complex for R , then $\varphi^! \omega_R$ is a dualizing complex for S .
- If R is F -finite then there exists a dualizing complex.

If dualizing complex are so abundant, how unique are they? As seen in the examples, they are not unique, but one can show that they are unique up to twisting with an invertible, i.e. up to $-\otimes_R^{\mathbb{L}} \mathcal{L}$ where $\mathcal{L} \in \mathcal{D}(R)$ is invertible.

In the case where $(R, \mathfrak{m})$ is local, there exist a criterion for a distinguished choice of a dualizing complex: in the given case there exists an isomorphism $\mathbb{R}\text{Hom}_R(R/\mathfrak{m}, \omega_R) \cong R/\mathfrak{m}[i]$. A dualizing complex ω_R is then called *normalized* if $i = 0$.

Lemma 4.26. *Let $(R, \mathfrak{m})$ be a local noetherian ring, $M \in \text{Mod}_R$ be a module of finite length, ω_R be a normalized dualizing complex, $E = E(R/\mathfrak{m})$ an injective hull of the residue field. Then*

$$\mathbb{R}\text{Hom}_R(M, \omega_R) \cong \text{Hom}_R(M, E).$$

Further, if $\dim(R) = 0$ then $\omega_R \cong E[0]$.

Proof. This can be proven by induction on the length of M using a short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow R/\mathfrak{m} \rightarrow 0$. The induction start is

$$\mathbb{R}\text{Hom}_R(R/\mathfrak{m}, \omega_R) \cong R/\mathfrak{m}[0] \cong \text{Hom}_R(R/\mathfrak{m}, E) \cong (E[\mathfrak{m}])[0].$$

$\square$

Lemma 4.27. *With the notation of the previous lemma*

$$\mathbb{R}\Gamma_{\mathfrak{m}} \omega_R \cong H_{\mathfrak{m}}^0(\omega_R) \cong E[0]$$

sketch. $R/\mathfrak{m}$ has finite length, so by the previous lemma

$$\mathbb{R}\text{Hom}_R(R/\mathfrak{m}^n, \omega_R) \cong \text{Hom}_R(R/\mathfrak{m}^n, E) \cong (E[\mathfrak{m}^n])[0].$$

Applying homotopy colimits yields an isomorphism

$$\mathbb{R}\Gamma_{\mathfrak{m}} \omega_R \cong E[0].$$

$\square$

We can now state and “prove” the local cohomology theorem.

Theorem 4.28. *Let $K \in \mathcal{D}(R)$, the rest of the notation as before. Then there exists an isomorphism*

$$\mathbb{R}\mathrm{Hom}_R(K, \omega_R)^{\wedge \mathfrak{m}} \cong \mathbb{R}\mathrm{Hom}_R(\mathbb{R}\Gamma_{\mathfrak{m}}K, E).$$

Proof. Using the previous lemmata this is now a simple calculation:

$$\begin{aligned} \mathbb{R}\mathrm{Hom}_R(K, \omega_R)^{\wedge \mathfrak{m}} &\cong \mathbb{R}\mathrm{Hom}_R(K^{\wedge \mathfrak{m}}, \omega_R^{\wedge \mathfrak{m}}) \\ &\cong \mathbb{R}\mathrm{Hom}_R(\mathbb{R}\Gamma_{\mathfrak{m}}K, \mathbb{R}\Gamma_{\mathfrak{m}}\omega_R) \\ &\cong \mathbb{R}\mathrm{Hom}_R(\mathbb{R}\Gamma_{\mathfrak{m}}K, E). \end{aligned}$$

□

What does this give us exactly? First of all we have a commutative diagram

$$\begin{array}{ccc} \mathcal{D}_{\mathrm{coh}}(R) & \xrightarrow{\mathbb{R}\Gamma_{\mathfrak{m}}} & \mathcal{D}_{\mathfrak{m}^{\infty}}(R) \\ \mathbb{R}\mathrm{Hom}_R(-, \omega_R) \downarrow & & \downarrow \mathrm{Hom}_R(-, E) \\ \mathcal{D}_{\mathrm{coh}}(R) & \xrightarrow{(-)^{\wedge \mathfrak{m}}} & \mathcal{D}_{\mathfrak{m}\text{-comp}}(R) \end{array}$$

If in particular R is complete, this reduces to

$$\begin{array}{ccc} \mathcal{D}_{\mathrm{coh}}(R) & \xrightarrow{\mathbb{R}\Gamma_{\mathfrak{m}}} & \mathcal{D}_{\mathrm{cof}}(R) \\ \mathbb{R}\mathrm{Hom}_R(-, \omega_R) \downarrow & \nearrow \mathbb{R}\mathrm{Hom}_R(-, E) & \\ \mathcal{D}_{\mathrm{coh}}(R) & & \end{array}$$

Here the vertical arrow and the diagonal arrows are equivalences, respectively the Grothendieck-Serre and Matlis duality. In particular the local cohomology theorem can be read as “local cohomology is the same as the composition of Grothendieck-Serre duality with Matlis duality”.

But we are still in a lecture for F -singularities, so how does the Frobenius come into play? Lets consider $(R, \mathfrak{m})$ a complete local noetherian ring of characteristic $p > 0$ and consider the above diagram again in the bounded setting

$$\begin{array}{ccc} \mathcal{D}_{\mathrm{coh}}^b(R) & \xrightarrow{\mathbb{R}\Gamma_{\mathfrak{m}}} & \mathcal{D}_{\mathrm{cof}}^b(R) \\ \mathbb{R}\mathrm{Hom}_R(-, \omega_R) \downarrow & \nearrow \mathbb{R}\mathrm{Hom}_R(-, E) & \\ \mathcal{D}_{\mathrm{coh}}^b(R) & & \end{array}$$

Lets consider what happens to $R \in \mathcal{D}_{\mathrm{coh}}^b(R)$ in the upper left corner. Under Grothendieck-Serre duality, R is mapped to the dualizing complex ω_R^2 . Further R is mapped under the horizontal arrow to $\mathbb{R}\Gamma_{\mathfrak{m}}R$. But we haven an explicit formula for this!

$$\mathbb{R}\Gamma_{\mathfrak{m}}R = \left(R \rightarrow \prod_i R_{f_i} \rightarrow \dots \rightarrow R_{f_1 \dots f_d} \right).$$

As already mentioned in example 4.2, we obtain from this diagram a natural map $\mathbb{R}\Gamma_{\mathfrak{m}}R \rightarrow F_*\mathbb{R}\Gamma_{\mathfrak{m}}R$.

If we now view R as an R -module with a Frobenius operation $R \xrightarrow{F} F_*R$, or equivalently

²Note that this equivalence is intrinsically derived, so we cannot calculate this term wise.

as a left $R[F] = R\{F\}/\langle r^p F - Fr \mid r \in R \rangle$ module, where $R\{F\}$ is the free non-commutative R -algebra on one generator, the formula for the local cohomology states, that the Frobenius action is compatible with $\mathbb{R}\Gamma_{\mathfrak{m}}$, so we can enhance the upper line as

$$\mathcal{D}_{\text{coh}}^b(R[F]) \xrightarrow{\mathbb{R}\Gamma_{\mathfrak{m}}} \mathcal{D}_{\text{cof}}^b(R[F]).$$

Can we do something similar for the downward maps? That is, can we say something about $\text{Hom}_R(-, E)$ with respect to the Frobenius? In fact, we can. Given a map $R \rightarrow F_*R$ we obtain a map

$$F_*E \cong F_*\text{Hom}_R(R, F^!E) \cong \text{Hom}_R(F_*R, E) \rightarrow \text{Hom}_R(R, E) \cong E.$$

Here we used in the first isomorphism that $F^!E$ is a injective hull, see exercise 4.4), so is non-canonically isomorphic to E . So $\text{Hom}_R(-, E)$ turns left $R[F]$ modules, also known as Frobenius modules, into right $R[F]$ modules, also known as Cartier modules. This looks great, at least at a first glance. However, the non-canoncity of the isomorphism $F^!E \rightarrow E$ is a bit problematic, especially as the right $R[F]$ module is given by the adjunction of this exact isomorphism.

In the meantime, this Problem was solved basically in the following two papers [[empty citation](#)] **Ich finde das Bhatt, Blickle, Tucker, Schwede paper nicht.....** and [MW24]. They proof that all the necessary choices can be made coherent, hence the diagram becomes

$$\begin{array}{ccc} \mathcal{D}_{\text{coh}, \text{Frob}}^b(R) & \xrightarrow{\mathbb{R}\Gamma_{\mathfrak{m}}} & \mathcal{D}_{\text{cof}, \text{Frob}}^b(R) \\ \mathbb{R}\text{Hom}_R(-, \omega_R) \downarrow & \nearrow \mathbb{R}\text{Hom}_R(-, E) & \\ \mathcal{D}_{\text{coh}, \text{Cartier}}^b(R) & & \end{array}$$

As a final touch of this course we will return to the notion of CM and use this as a motivation for introducing perverse sheaves, not using the perverse t -structure, but an explicit definition.

Remark 4.29. Let $(R, \mathfrak{m})$ be a local noetherian ring. Then R is CM if and only if $\mathbb{R}\Gamma_{\mathfrak{m}}R \cong H_{\mathfrak{m}}^d R[-d]$ where d is the dimension of R . By re-shifting we get that this is equivalent to $\mathbb{R}\Gamma_{\mathfrak{m}}R[d] \cong H_{\mathfrak{m}}^d(R)$ which in turn is equivalent to $\omega_R[d] \cong H^{-d}\omega_R$.

In summary, a local noetherian ring R is CM if and only if the normalized dualizing complex is concentrated in degree $d = \dim(R)$, or $\mathbb{D}(R[d]) = \mathbb{R}\text{Hom}_R(R[d], \omega_R)$ is an R -module concentrated in degree 0.

Definition 4.30. Let $M \in \mathcal{D}_{\text{coh}}^b(R)$. Then M is called a *perverse sheave* by definition if and only if $\mathbb{D}(M)$ is an R module concentrated in degree 0. In particular, R is CM if and only if $R[d]$ is perverse.

We can now apply this definition to F -rationality to obtain a different viewpoint on this type of singularities.

Remark 4.31. Let $(R, \mathfrak{m})$ be a local noetherian ring of characteristic $p > 0$. Then R has F -rational singularities by definition if and only if R is CM and the composition $F^e : H_{\mathfrak{m}}^i R \xrightarrow{F_*^e \circ F^e} F_*^e H_{\mathfrak{m}}^i(R)$ is injective for all e and all $c \in R \setminus R^0$. By Matlis duality the second condition is equivalent to $F_*^e \omega_R \xrightarrow{c} F_*^e \omega_R \rightarrow \omega_R$ is surjective.

The first condition can be rewritten using local cohomology as $H_{\mathfrak{m}}^d(R) \supseteq 0_{H_{\mathfrak{m}}^d}^*(R) = 0$. But by a result of Karen Smith, $0_{H_{\mathfrak{m}}^d}^*(R)$ is the maximal nontrivial $R[F]$ submodule in $H_{\mathfrak{m}}^d(R)$, so R is CM if and only if $H_{\mathfrak{m}}^d(R)$ is a simple left $R[F]$ -module, which in turn is equivalent to $H^{-d}\omega_R$ is a simple right $R[F]$ -module.

In Summary, R is F -rational if and only if $H^{-d}\omega_R$ is a simple right $R[F]$ -module and the map $F_*^e \omega_R \xrightarrow{c} F_*^e \omega_R \rightarrow \omega_R$ is surjective for all e and $c \in R \setminus R^0$.

With this remark at hand we can compare the notion of F -rationality with the general notion of a rational singularity, or, to be more precise, pseudo-rationality.

Definition 4.32. Let R be a ring. Then R is called *pseudo-rational* if R is CM and for all proper birational maps of schemes $X \xrightarrow{\pi} \text{spec}(R)$ the trace map $\text{Tr}_\pi : \pi_*\omega_X \rightarrow \omega_R$ is surjective. here the trace map is the map induced from $R \rightarrow \pi_*\mathcal{O}_X$ under the Grothendieck-Serre duality.

If everything happens over a field of characteristic 0 then R is pseudo rational if and only if for one and hence all resolutions of singularities $X \rightarrow \text{spec}(R)$ the trace map is surjective.

Proposition 4.33. *F -rationality implies pseudo rationality.*

Proof. By naturality of Tr_π , the trace map is a map of right $R[F]$ -modules. In particular, $\text{Im}\text{Tr}_\pi \subseteq \omega_R$ is a right $R[F]$ submodule. So by the above remark

$$\text{Im}\text{Tr}_\pi = \begin{cases} \omega_R \\ 0 \end{cases}$$

But the trace is compatible with localisation, and as one can check easily, the trace is generically the identity. In particular the image is non zero, proving the claim. $\square$

One can show that the weaker notion of *alteration pseudo-rational*, where one allows for alteration maps instead of only proper birational maps, is equivalent to F -rational.

What could one study next? let us return to the local duality diagram, then we can extend this to a diagram on the left as follows where $X = \text{spec}(R)$.

$$\begin{array}{ccc} \mathcal{D}_C^b(X_{\text{ét}}, \mathbb{F}_p) & \xleftarrow{(-)^{Frob=1}} & \mathcal{D}_{\text{coh}, \text{Frobenius}}^b(R) \\ \mathbb{R}\text{Hom}_R(-, \omega_R)^{Frob=1} \uparrow & & \nwarrow \mathbb{R}\text{Hom}_R(-, \omega_R) \\ \mathcal{D}_{\text{coh}, \text{Cartier}}^b(R) & & \end{array}$$

Here the upper horizontal map is given by Frobenius fix points. The map was studied By Bhatt and Lurie in their paper on the Reimann-Hilbert Correspondence. It turns out that this map is surjective but loses exactly the Modules on which the Frobenius acts nilpotent. From this one learns the following:

A ring R is CM if and only if $R[d]$ is a perverse $R[F]$ module which implies that $\mathbb{F}_p[d]$ is a perverse $\mathbb{F}_p$ sheaf on $X_{\text{ét}}$. Further R is CM up to nilpotent Frobenius operation if and only if $\mathbb{F}_p[d]$ is a perverse $\mathbb{F}_p$ sheaf on $X_{\text{ét}}$.

In particular R is F -rational implies that $\mathbb{F}_p[d]$ is a simple perverse $\mathbb{F}_p$ sheaf.

Bibliography

- [AM18] M. F. Atiyah and I. G. MacDonald. *Introduction to commutative algebra*. CRC Press, 2018.
- [BS15] Bhargav Bhatt and Peter Scholze. “Projectivity of the Witt vector affine Grassmannian”. In: *arXiv e-prints*, arXiv:1507.06490 (July 2015), arXiv:1507.06490. DOI: 10.48550/arXiv.1507.06490. arXiv: 1507.06490 [math.AG].
- [Har13] Robin Hartshorne. *Algebraic geometry*. Vol. 52. Springer Science & Business Media, 2013.
- [Mat89] Hideyuki Matsumura. *Commutative ring theory*. 8. Cambridge university press, 1989.
- [MP21] Linquan Ma and Thomas Polstra. “F-singularities: a commutative algebra approach”. In: *Preprint available at <https://www.math.purdue.edu/ma326/F-singularitiesBook.pdf>* 2 (2021).
- [MW24] Klaus Mattis and Timo Weiß. *The derived ∞ -category of Cartier Modules*. 2024. arXiv: 2410.17102 [math.AG]. URL: <https://arxiv.org/abs/2410.17102>.
- [Sta24] The Stacks project authors. *The Stacks project*. <https://stacks.math.columbia.edu>. 2024.
- [Wei94] Charles A Weibel. *An introduction to homological algebra*. 38. Cambridge university press, 1994.